BUSINESS INTELLIGENCE AND DATA ANALYTICS

University of Petra

Institute of Analytics
Recognized Programme

307102
Descriptive Statistics for Business

Introduction to Inferential Statistics

2024-1

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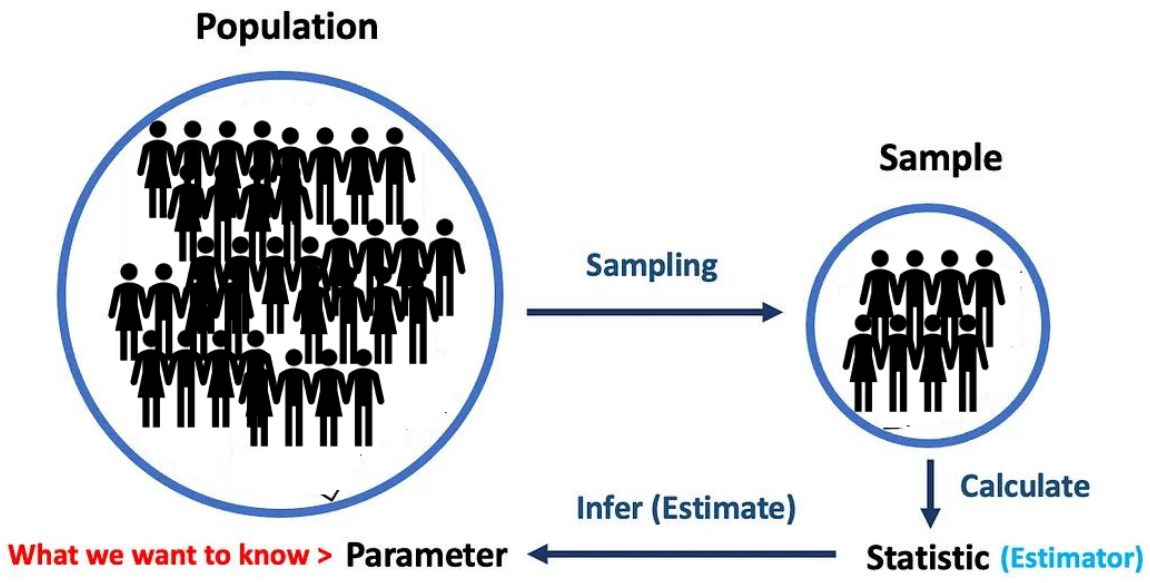
Introduction to Inferential Statistics

- **Definition**: Inferential statistics is the branch of statistics that involves making inferences or predictions about a population based on sample data.
- **Objective**: The main goal of inferential statistics is to draw conclusions, make predictions, or test hypotheses about a population using information obtained from a sample.

Introduction to Inferential Statistics

Key Concepts:

- **Population**: The entire group of individuals or items of interest that we want to study.
- **Sample**: A subset of the population from which data is collected and analyzed.
- **Parameter**: A numerical characteristic of a population (e.g., mean, proportion).
- **Statistic**: A numerical characteristic of a sample (e.g., sample mean, sample proportion).



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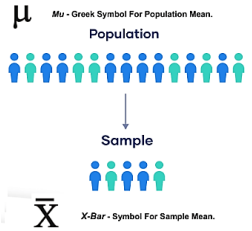

Types of Inferential Statistics

1. Estimation
   - Point Estimation: Provides a single best guess about a population parameter (e.g., the mean, proportion).
   - Interval Estimation (Confidence Intervals): Gives a range of values for a population parameter with a certain level of confidence.
2. Hypothesis Testing
   - A statistical method that uses sample data to evaluate a hypothesis about a population parameter, determining whether there is sufficient evidence to accept or reject a given hypothesis about that population.
3. Regression Analysis
   - Assesses the linear relationship between one continuous dependent variable and one or more independent variables.

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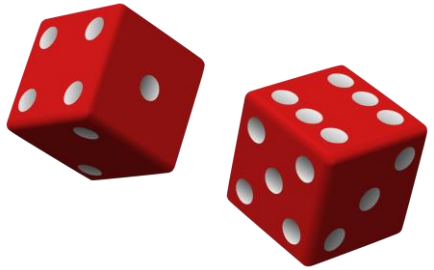
Introduction to Random Variables

Understanding the Basics of Probability in Statistics

What is a Random Variable?



- A random variable is a variable whose possible values are outcomes of a random process.
- A random process is a process or event or where the outcome cannot be predicted with certainty, and each possible outcome has an associated probability.
- It can be classified as discrete or continuous depending on the range of values that it assumes.
- A discrete random variable assumes a countable number of distinct values, whereas a continuous random variable is characterized by measurable (uncountable) values.
- Example: Flipping a coin (discrete), measuring the amount of rain in a day (continuous)

Discrete Random Variables

A Discrete Random Variable is a variable which can take on only a countable number of distinct values like 0, 1, 2, 3, 4, 5…100, 1 million, etc.

Examples of discrete random variables:

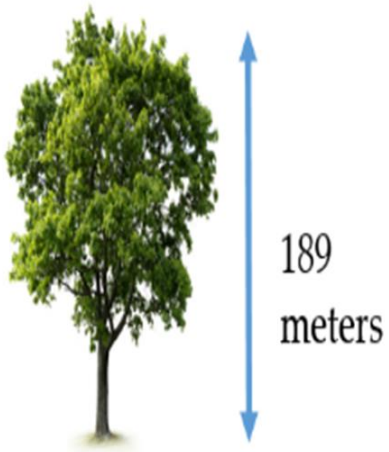
- The number of cars that passes through a toll booth everyday in a given day.
- The number of times a coin lands on tails after being flipped 20 times.
- The number of defective widgets in a box of 50 widgets.
- Number of Calls Received by a Call Center.
- Number of Emails Received in a Day

Continuous Random Variables

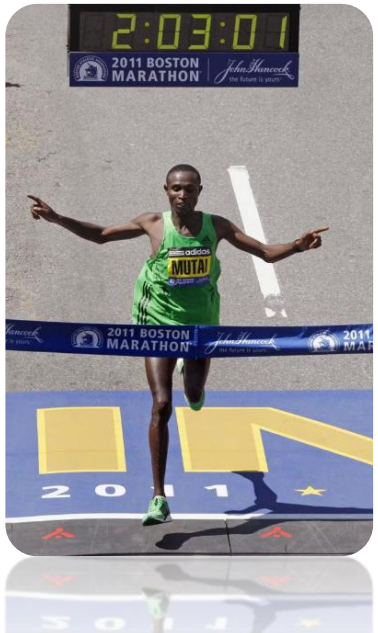
A Continuous Random Variable is a variable which can take on an infinite number of possible values.

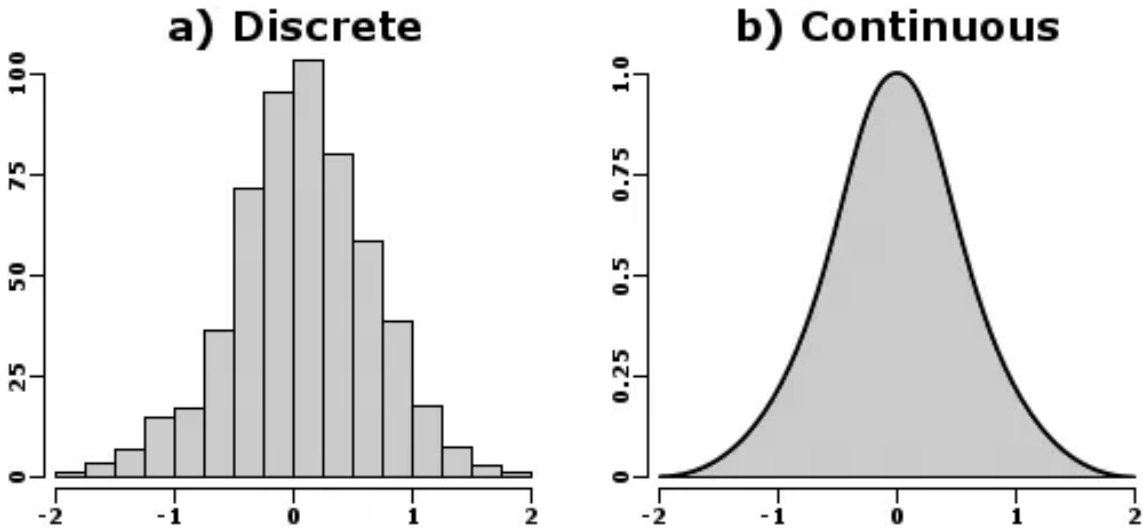
Examples of continuous random variables:

- Business profits.
- The height of a tree.
- The weight of an animal.
- Arrival times of airlines.
- Time required to run a marathon.
- The weight of a chocolate bar on a production line.



Tree A





Introduction to Probability Distributions

Discrete and Continuous Probability Distributions

Introduction to Probability Distributions

- Probability distributions are mathematical functions that describe the likelihood of various outcomes in a statistical experiment.
- They provide a mathematical framework for modeling random phenomena.

Discrete vs. Continuous Probability Distributions

- Discrete distributions: Outcomes are countable and have finite or countably infinite values.
- Continuous distributions: Outcomes are uncountable and have an infinite range of values.

- For a probability distribution to be valid, it must satisfy the following two criteria:
  - The probability for each outcome must be between 0 and 1.
  - The sum of all the probabilities must add up to 1.

Discrete Probability Distributions
The Binomial Distribution

- Models the number of successes in a fixed number of independent Bernoulli trials.
- Bernoulli trial is the experiment that has only two outcomes (Yes and No).
- Parameters: n (number of trials) and p (probability of success on each trial).

$$P_x = \binom{n}{x} p^x q^{n-x}$$

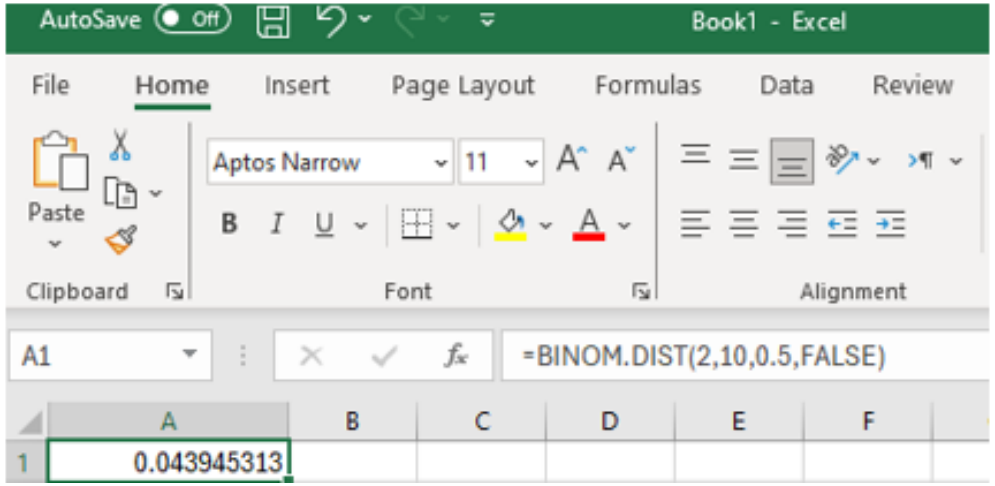
P	= binomial probability
x	= number of times for a specific outcome within n trials
$\binom{n}{x}$	= number of combinations
p	= probability of success on a single trial
q	= probability of failure on a single trial
n	= number of trials

Example: The defect ratio in a car factory is 2% and the factory produces 80 cars in a month,. We can use the binomial distribution to answer questions similar to:

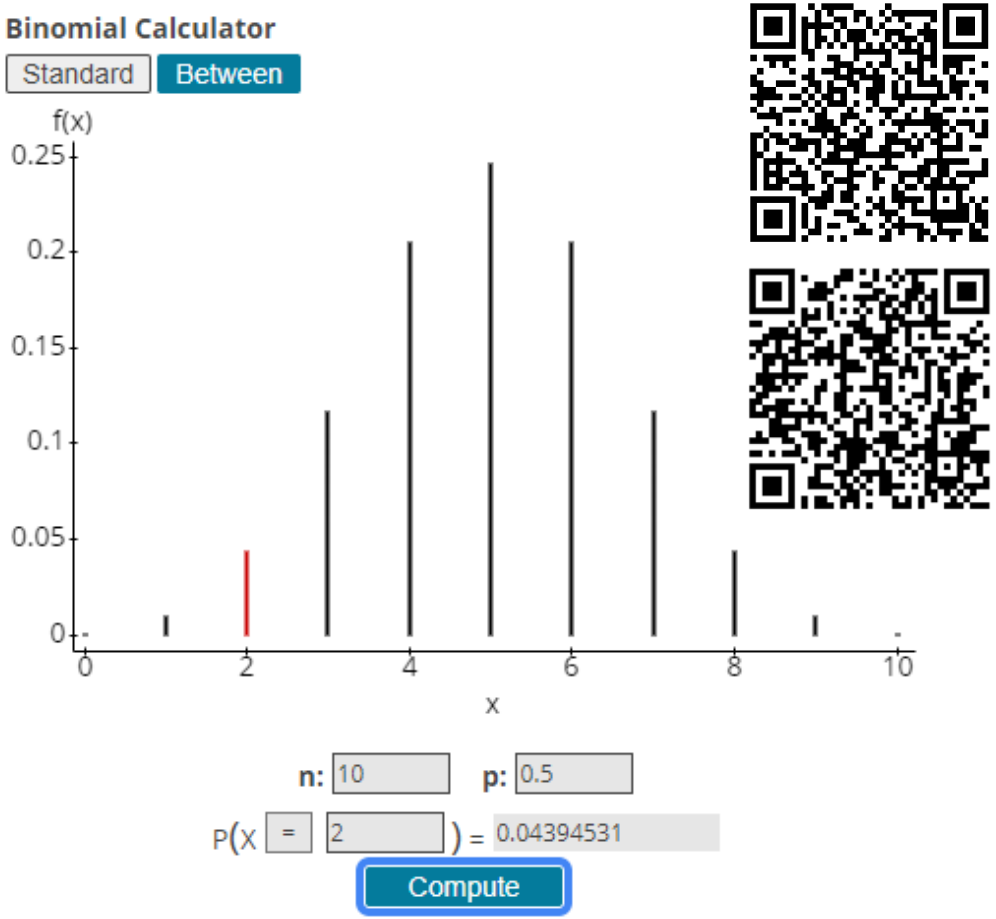
- what is the probability that exactly 2 cars are defective?
- what is the probability that less than 2 cars are defective?
- what is the probability that more than 2 cars are defective?
- What is the probability that all cars produced in a month will have no defects?

The Binomial Distribution

What is the probability of having 2 Heads ONLY out of 10 trials, knowing that the probability of Head = 50%?



Excel Function
=BINOM.DIST(2,10,0.5,FALSE)



The larger the number of trials (e.g. ask 100 students) the more the distribution will look like normal distribution.

Discrete Probability Distributions
The Poisson Distribution

- Poisson Distribution can be used to model the number of times an event occurs within a specified time period or in a given space.
- It's particularly useful for estimating the likelihood of rare events over short intervals.
- The key feature of the Poisson distribution is that it is defined by just one parameter, **λ** (lambda), which represents the average rate at which events occur.

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

where

x = 0, 1, 2, 3, ...

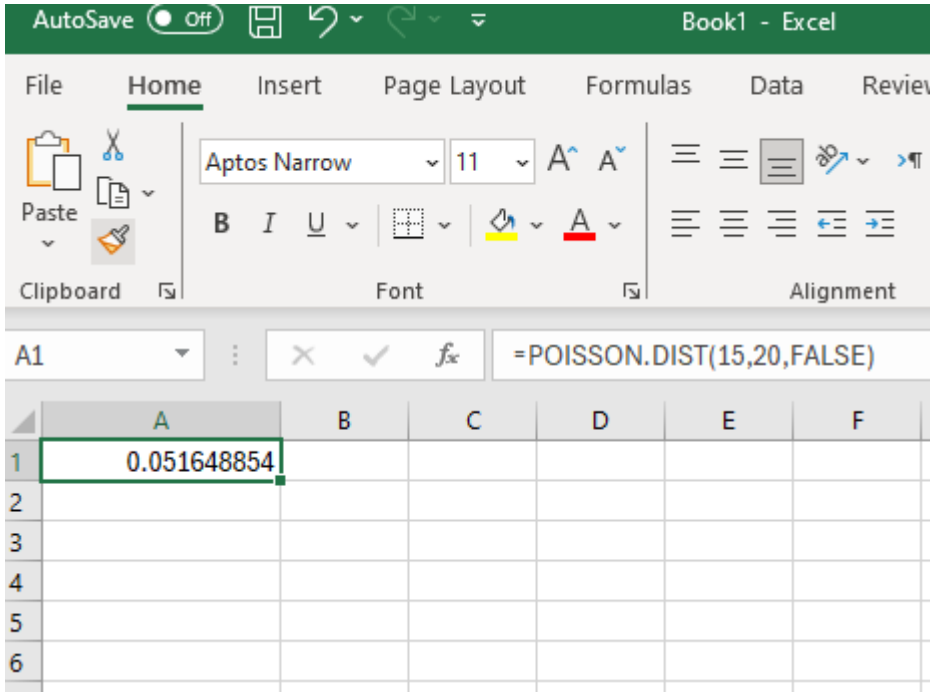
λ = mean number of occurrences in the interval

e = Euler's constant $\approx$ 2.71828

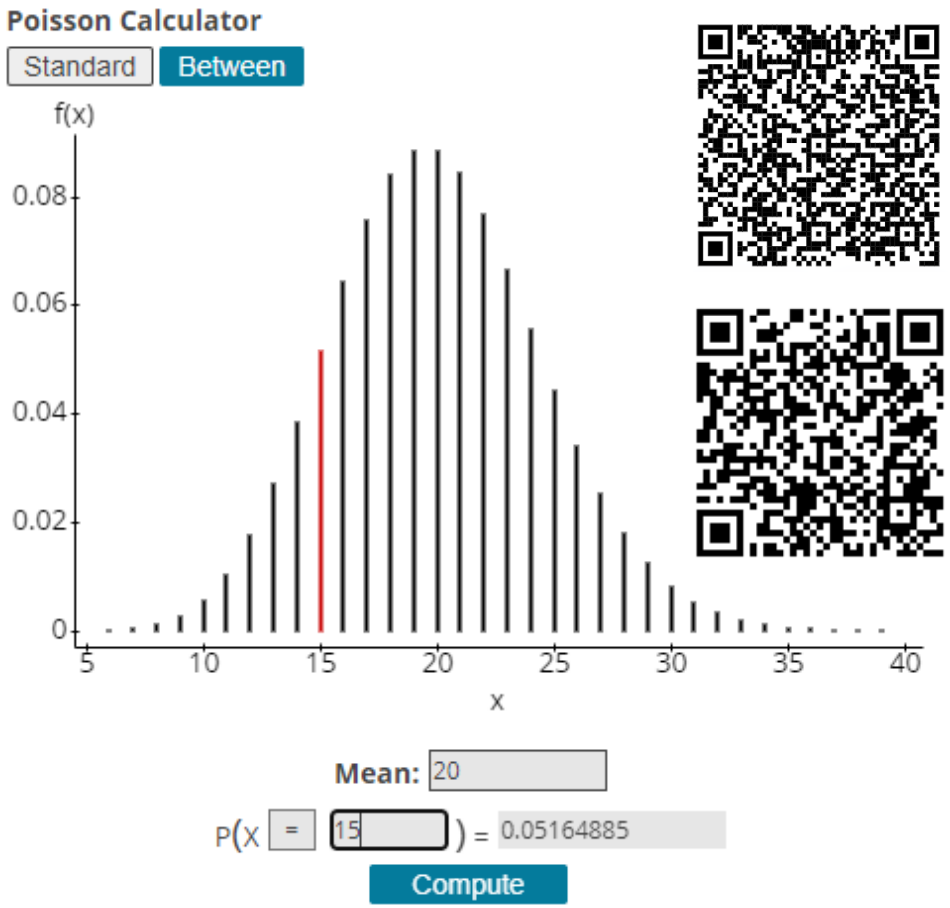
Example: Suppose that Zein call center receives 1000 calls in a single hour, we can use the Poisson distribution to answer questions similar to:

- What is the probability that the call center receives more than 1100 calls in an hour?
- What is the probability that the number of calls will exceed the capacity of the call center's current staffing levels, requiring additional temporary staff?

Poisson Distribution



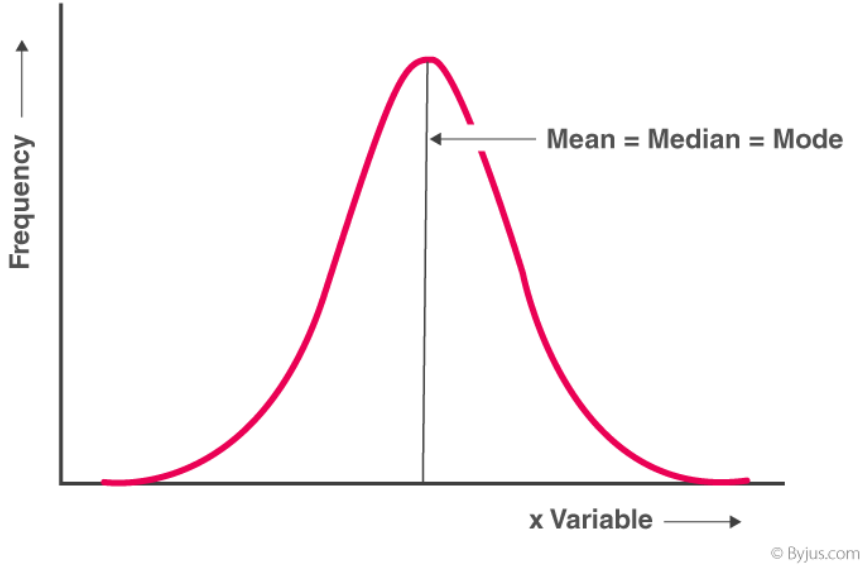
Excel Function
=POISSON.DIST(15,20,FALSE)



The larger the average rate **λ,** the more the distribution will look like normal distribution.

Continuous Probability Distribution
The Normal Distribution

- The normal distribution is the most extensively used probability distribution in statistical work.
- It is also known as the Gaussian distribution and the Bell Curve.
- The x-axis represents the outcomes of the normal process (e.g. students scores, heights, salaries…etc.).
- The AREA UNDER THE CURVE represents the probability of the outcome.
- The normal distribution has a symmetric distribution where most of the observations cluster around the central peak.
- Theoretically, a normal random variable can assume any value between minus infinity and plus infinity.
- Normal Distribution Main Features:
  - Symmetrically distributed
  - Long Tails / Bell Shaped
  - Mean/ Mode and Median are the same



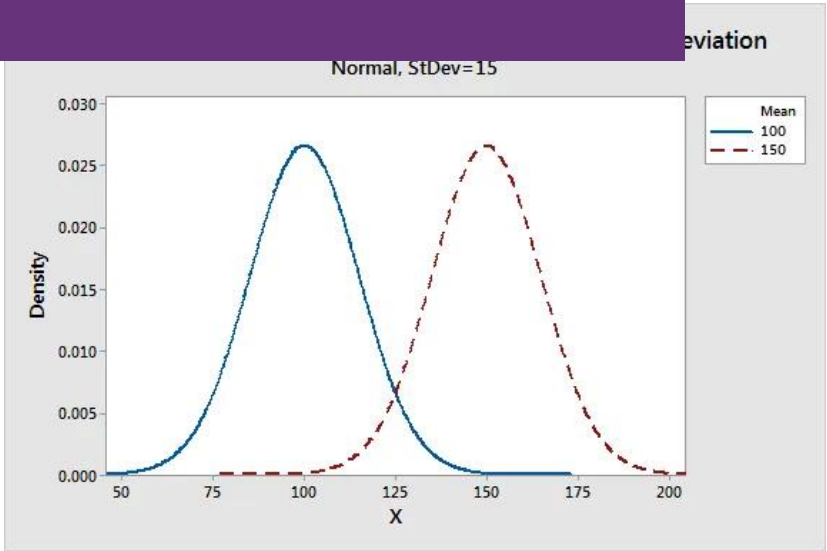
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

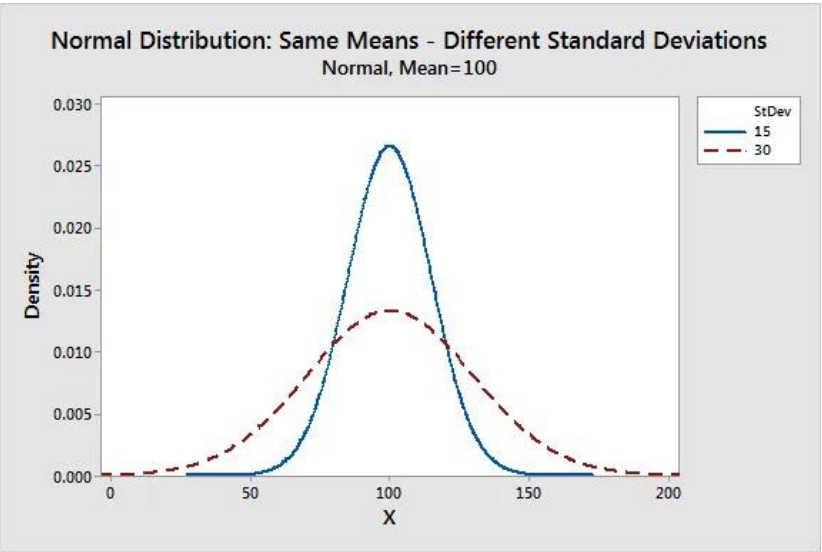
Parameters of the Normal Distribution

- The normal distribution is completely described by two parameters, the Mean and the Standard Deviation.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

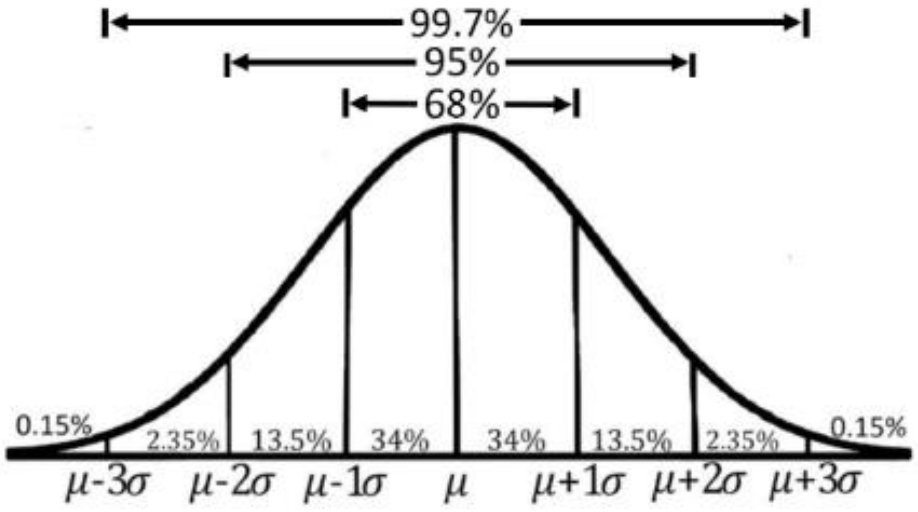
- Where X is a continuous random variable (the expected outcome of a continuous random process).
- μ = The Mean of the Outcomes of the Random Process.
- σ = The Standard Deviation of the Outcomes of the Random Process.
- π is approximately 3.14159.
- e is approximately 2.71828.





The Empirical Rule for the Normal Distribution

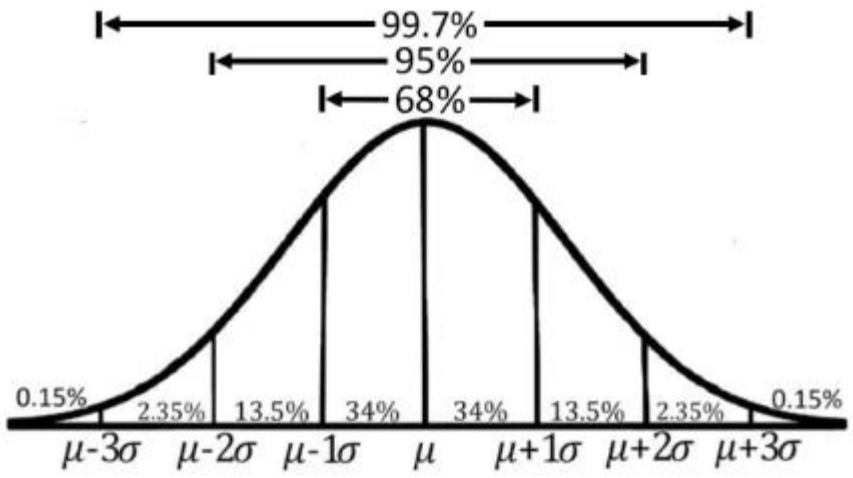
- The Empirical Rule describes how data (continuous outcomes) are spread in a normal distribution.
- The proportion of the area that falls under the curve between two points on a probability distribution indicates the probability that a value will fall within that interval.
- If we know that the data are drawn from a relatively symmetric and bell-shaped distribution - perhaps by a visual inspection of its histogram, then we can make statements about the percentage of observations that fall within certain intervals.



- Approximately 68% of data fall within one standard deviation (σ) of the mean (μ).
- Around 95% are within two standard deviations.
- About 99.7% lie within three standard deviations.

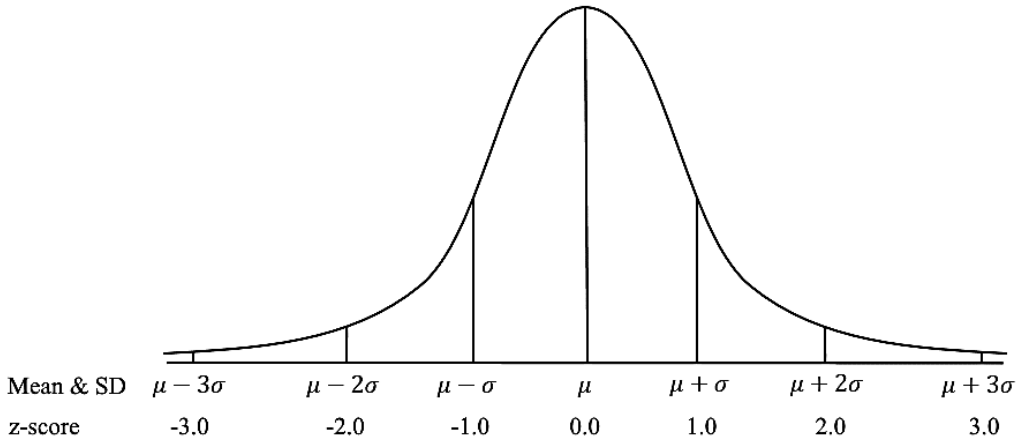
Empirical Rule Example

- A large lecture class has 280 students. The professor has announced that the mean score in an exam is 74 with a standard deviation of 8.
- Use the Empirical Rule to answer the following:
  a. Approximately how many students within 58 and 90?
  b. Approximately how many students scored higher than 90?



The Z - Score

- Definition: Z-Score represents the number of standard deviations a data point is from the mean of its distribution.
- Formula: $Z = \frac{x - \mu}{\sigma}$



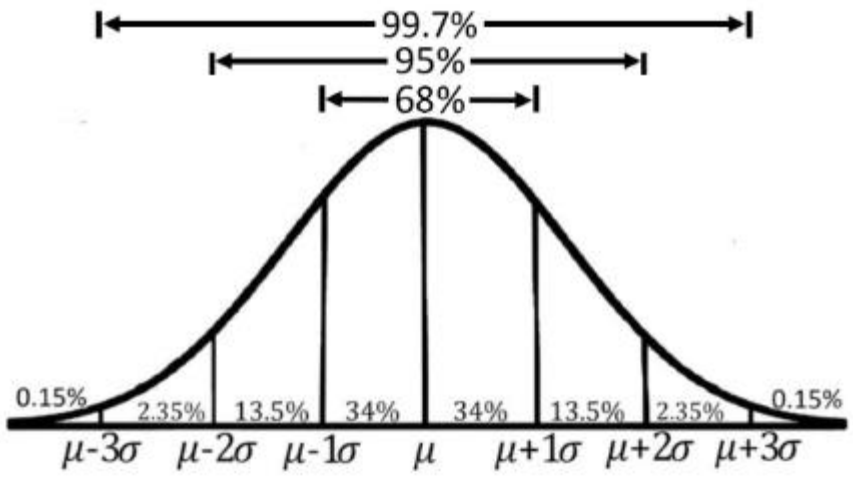
Where X is a data point, μ is the mean, and σ is the standard deviation.

Interpretation:

- A z-score of 0 corresponds to a data point that is exactly at the mean.
- Positive z-scores indicate values above the mean.
- Negative z-scores indicate values below the mean.

Back to our Empirical Rule Example

- A large lecture class has 280 students. The professor has announced that the mean score in an exam is 74 with a standard deviation of 8.
- Use the Empirical Rule to answer the following:
  a. Approximately how many students within 58 and 90?
  b. Approximately how many students scored higher than 90?

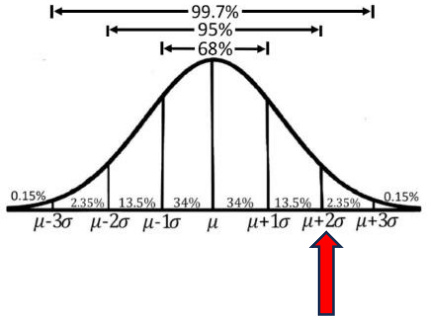


Solution for part (a)

- First, we find the z-scores corresponding to 58 and 90.
- Z for 58 = (58 – 74) / 8 = -16/8 = -2, Z for 90 = (58 – 74) / 8 = 16/8 = +2
- Using the empirical rule, we know that 95% of observations falls within 2 standard deviations (each side) from the mean, therefore, the number of students under this criteria = 280 x .95 = 266 students.

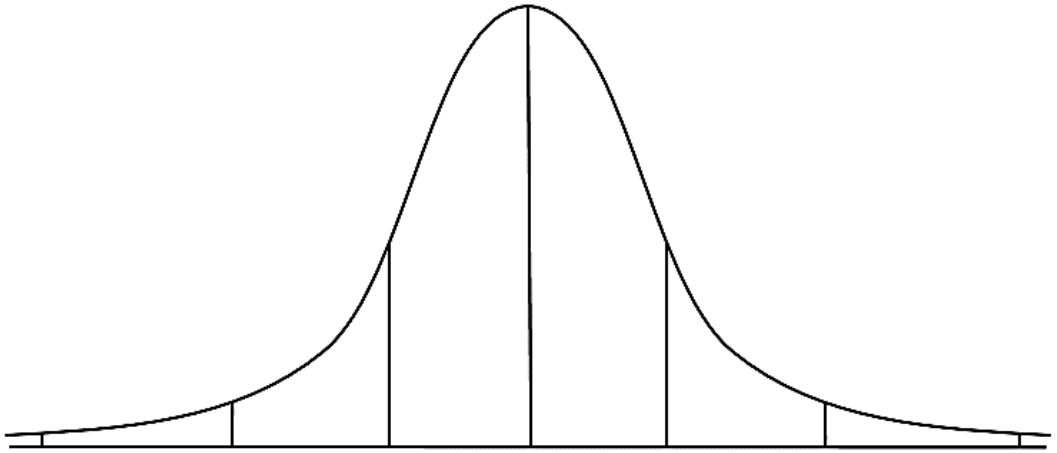
Solution for part (b)

- The score 90 is 2 standard deviations above the mean.
- Therefore, 97.5% of students are below that value and 2.5% are above it.
- Consequently, 2.5/100 * 280 = 7 students are above 90.



Z-Scores Uses in Statistics

- **Standardization**: Z-scores standardize different distributions to be comparable by converting them to a common scale.
- **Identification of Outliers**: Data points with z-scores less than -2 or greater than 2 are often considered outliers.
- **Compare values** from different populations (e.g. exam results between courses).
- **Probabilities and Percentiles**: Z-scores link data points to their probabilities and percentiles in a r o the 97.5th percentile.

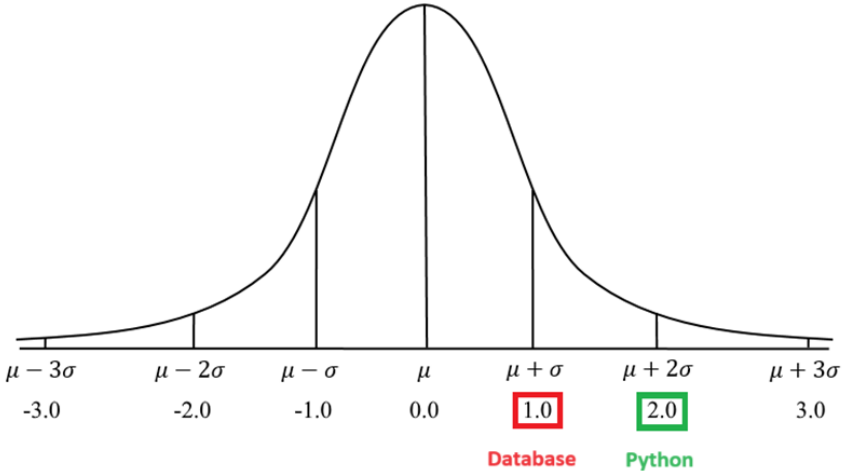


Mean & SD	$\mu - 3\sigma$	$\mu - 2\sigma$	$\mu - \sigma$	μ	$\mu + \sigma$	$\mu + 2\sigma$	$\mu + 3\sigma$
z-score	-3.0	-2.0	-1.0	0.0	1.0	2.0	3.0
Percentile	0.1%	2.3%	15.9%	50.0%	84.1%	97.7%	99.9%

Z-Scores Example – Comparing Values

- Khalid achieved 90 in both Python & Database exams. The mean μ for the Python exam was 70 and σ = 10, The μ for the Database exam was 80 and σ = 10.
- Use standard scores to determine where did Khalid achieve a better score.

- Standard score for the Python Exam: (90−70)/10 =2
- Standard score for the Database Exam: (90−80)/10 =1
- Khalid's Z-score for the Python exam is 2, which means his score was 2 standard deviations above the mean.
- His Z-score for the Database exam is 1, indicating his score was 1 standard deviation above the mean.
- Khalid performed better on the Python exam than Database exam since his Z-score is higher for that exam.



Z-Tables

The Z-Tables

- The area under the curve in continuous probability distributions represents the probability of outcome occurrence.
- The Empirical Rule has <u>pre-defined</u> areas for certain Z-Scores.
- What can we do if we need the area between other Z-Scores e.g. +1.25, -2.33?
- It will be hard to compute the integral for different intervals:

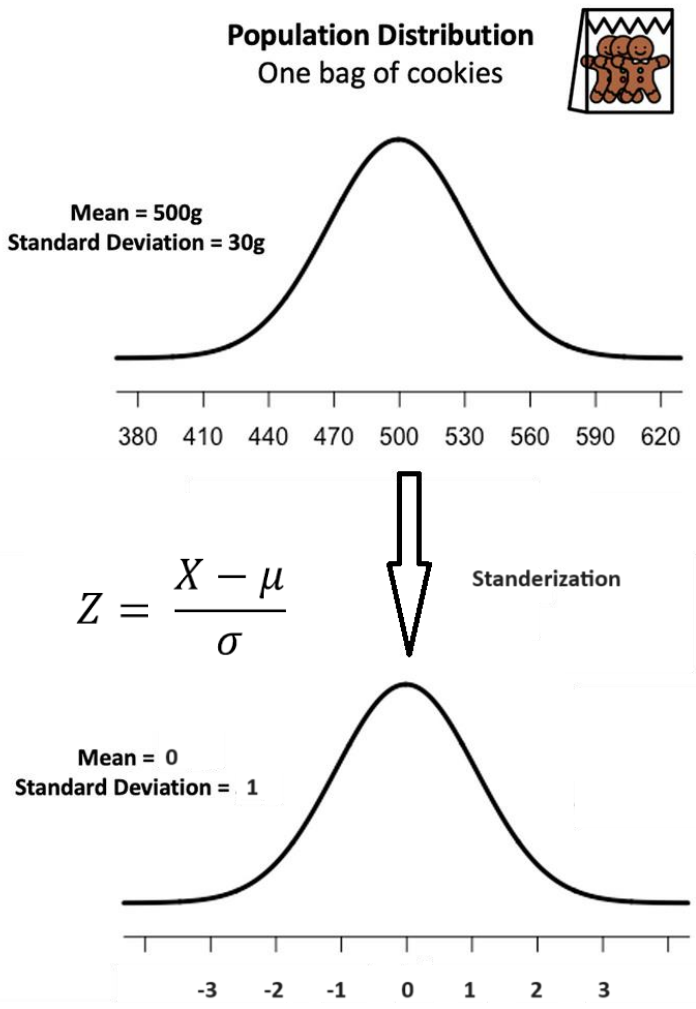
$$P(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \, dx$$

- A simp ... table of pre-co
- But we need to compute that area ONCE. Therefore, we need to create a Standard Distribution.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Standard Normal Distribution and Standard Z Scores

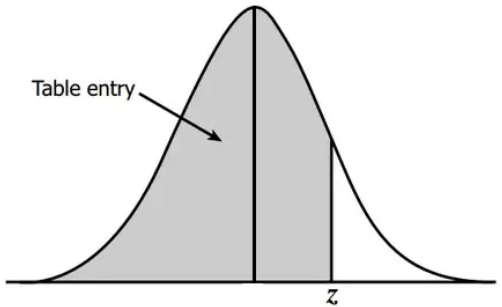
- The standard normal distribution is a special case of the normal distribution where the mean is zero and the standard deviation is 1.
- This distribution is also known as the Z-distribution.
- A value on the standard normal distribution is known as a standard score or a Z-score.
- A standard score represents the number of standard deviations above or below the mean that a specific observation falls.
- For example, a standard score of 1.5 indicates that the observation is 1.5 standard deviations above the mean.
- On the other hand, a negative score represents a value below the average.
- The mean has a Z-score of 0.



https://medium.com/towards-data-science/what-is-p-value-370056b8244d

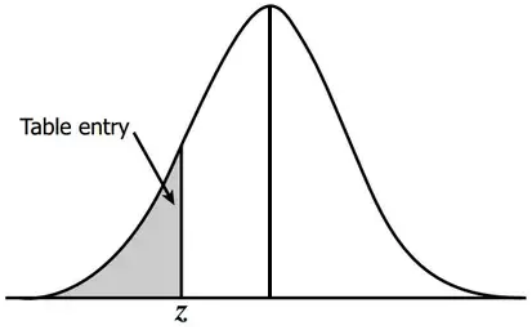
Z-Table for Z Values above 0

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998



Z-Table for Z Values below 0

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
–3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
–2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
–2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
–2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
–2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
–2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
–1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
–1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
–1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
–1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
–1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
–1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
–1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
–1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
–1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
–1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
–0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
–0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
–0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
–0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
–0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
–0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641



Z-Tables Example

- The average score for the MID exam in Descriptive Statistics course was 70. The standard Deviation for that exam was 6.
- Knowing that the exam scores are normally distributed, what is the probability of having a score higher than 75 in that exam?

First, we need to find the standard position (z-score) for the 75 score.
Z score for (75) = (75 – 70) / 6 = 5/6 = .833
Using the z table, the closest value for .833 is: 0.7967 which corresponds to 79.67% of scores are below 75.
To find the probability of having a score over 75, we subtract 79.67% out of the total probability of 100%:
100 – 79.67 = 20.33%

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
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0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
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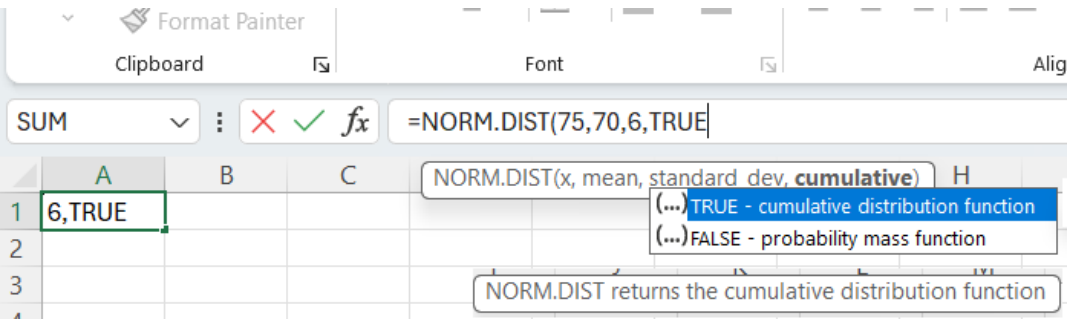
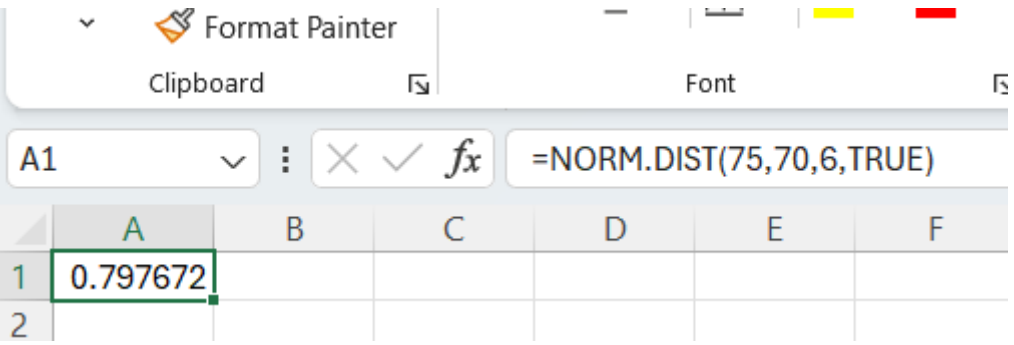
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Z-Tables using Excel Functions

- The average score for the MID exam in Descriptive Statistics course was 70. The standard Deviation for that exam was 6.
- What is the probability of having a score higher than 90 in that exam?
- We can use Excel NORM.DIST function. This function expects 4 values, the value we want to find its probability, the mean, the standard deviation and we set cumulative = TRUE.
- NORM.DIST(75,70,6,True) = 0.797672
- Therefore, the probability of having a score > 75 = 1 - .797672 = .2023% ~ 20%





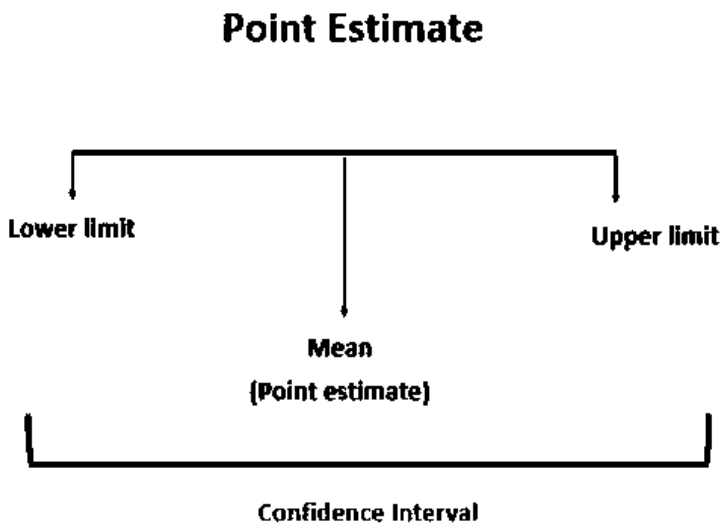
Estimation in Statistics

Estimation in Statistics

- Often in statistics we're interested in measuring population parameters.
- Two of the <u>most common</u> population parameters are:
  1. Population mean: the mean value of some variable in a population (e.g. the mean height of males in the U.S.)
  2. Population proportion: the proportion of some variable in a population (e.g. the proportion of residents in a county who support a certain law)
- Although we're interested in measuring these parameters, it's usually too costly and time-consuming to go around and collect data on every individual in a population in order to calculate the population parameter.
- Instead, **<u>we typically take a random sample</u>** from the overall population and use data from the sample to **<u>estimate</u>** the population parameter.

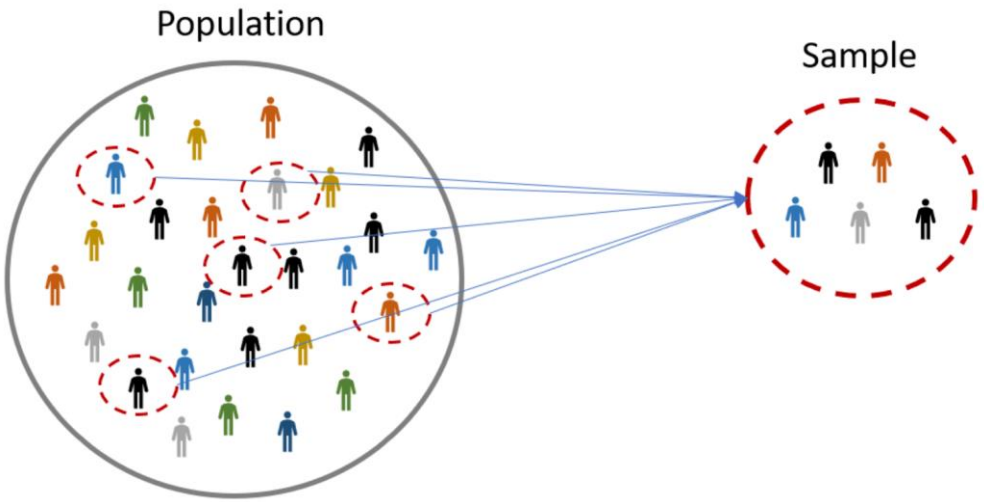
Point vs. Interval

- A point estimate is a single value estimate of a parameter. For instance, a sample mean is a point estimate of a population mean.
- An interval estimate gives you a range of values where the parameter is expected to lie.
- Both types of estimates are important for gathering a clear idea of where a parameter is likely to lie



Point Estimates

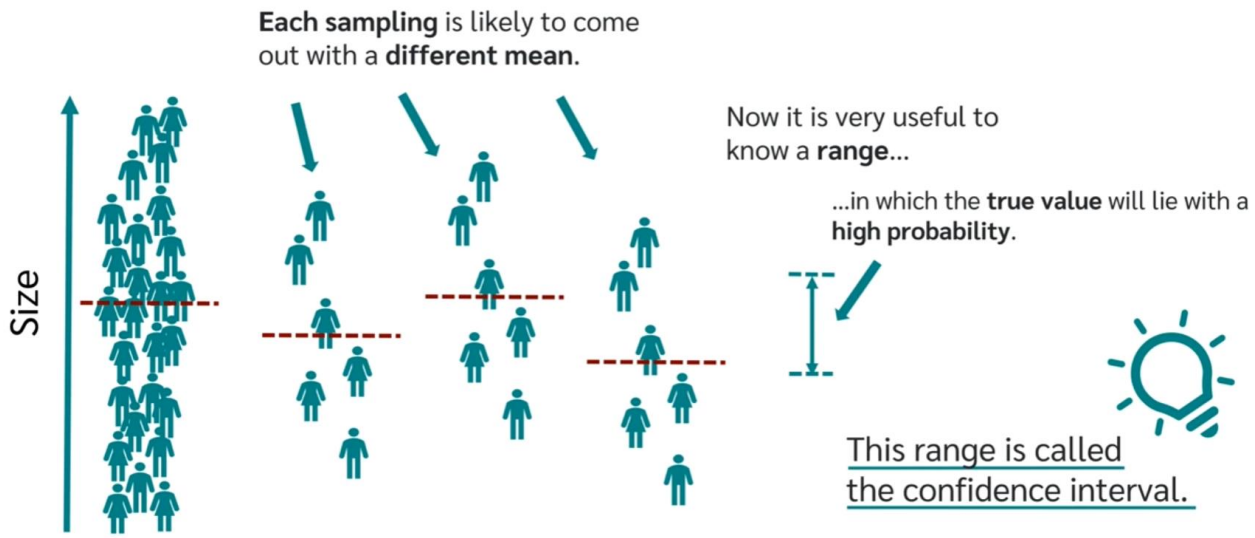
- Suppose we want to estimate the mean weight of a students in the university.
- Since there are thousands of students in the university, it would be extremely time-consuming and costly to go around and weigh each individual student.
- Instead, we might take a random sample of 50 students and use the mean weight of the students in this sample to estimate the true population mean.



- In this case, the mean weight of the sample is called a Point Estimate for the true mean weight of the population.

Interval Estimates - Confidence Intervals

- The problem with the previous example is that the mean weight of students in the sample **is not guaranteed** to exactly match the mean weight of students in the whole population.
- For example, we might just happen to pick a sample full of low-weight students or perhaps a sample full of heavy students.
- To capture this uncertainty, we can create an Interval Estimate or a confidence interval around our point estimate.
- The confidence interval gives us a range of values that are likely to contain the true population parameter.



Computing Confidence Intervals

To compute the confidence interval for the mean of a normally distributed data we use the formula below:

Confidence Interval = (point estimate) ± Margin of Error

Confidence Interval = (point estimate) ± [(critical value)*(standard error)]

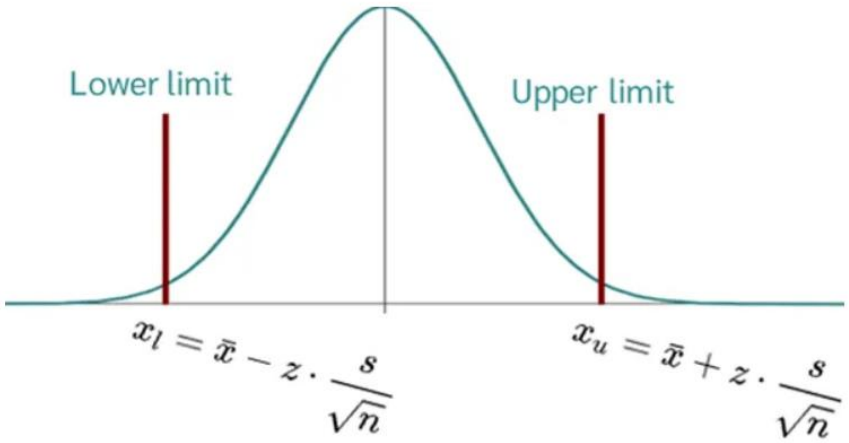
The critical value is the z-score and the standard error = $s/\sqrt{n}$

Confidence Interval is calculated as:

Point Estimate ± Margin of Error

Point Estimate ± (Critical Value) (Standard Error)

$$\text{C.I.} = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$



https://medium.com/@ashisharora2204/hypothesis-testing-confidence-interval-level-margin-of-error-39aa7c7ddcd2

The Significance Level Alpha (α) and The Confidence Level

- To compute the confidence interval, we need to define the Significance Level Alpha (α) or the Confidence Level.
- The two terms are related to each other according to the following equation:

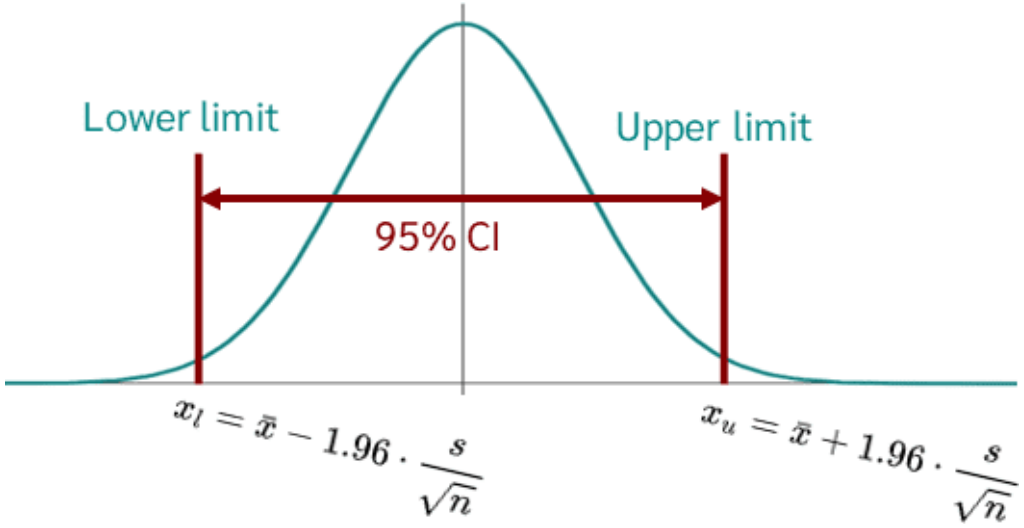
  Confidence level= 1 – α

- Common values for these notations are listed in the table

Alpha - α	**5%**	**1%**
Confidence level	**95%**	**99%**
z-Value	**1.96**	**2.58**

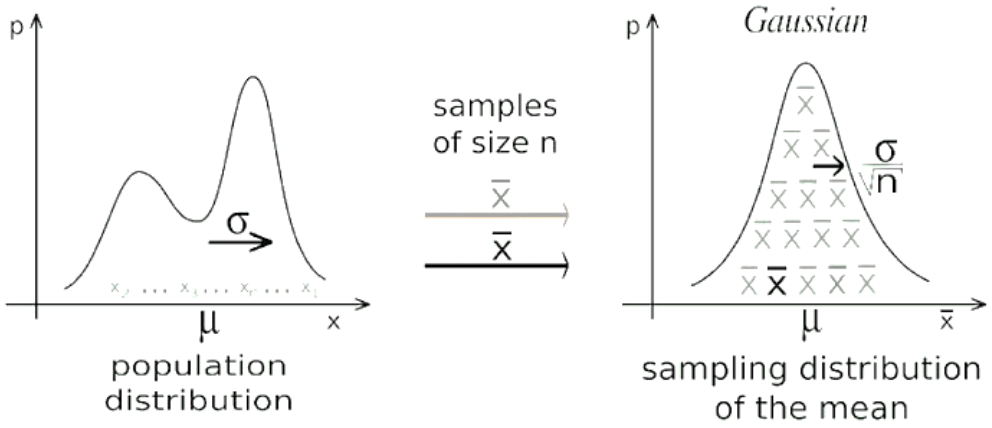
The Significance Level Alpha (α) and The Confidence Level

- The confidence level (e.g. 95%) means that if we collected 100 samples and created confidence intervals for each of these samples, we would expect that 95% of these confidence intervals will contain the true population parameter.
- It's important to note that this is a theoretical long-term proportion—it's about what we'd expect to happen over many repetitions of the same process, not a guarantee for any individual interval.



Where did the Standard Error Come From?
The Central Limit Theorem

- The Central Limit Theorem (CLT) states that samples means created from large number of samples will be approximately normally distribution although the distribution we sample from might not be normal.
- When we collect large number of samples and we compute the average for each of these samples, we call the generated means as the Sampling Distribution of the means.
- The distribution of samples means becomes more normal when the samples size increases.



According to CLT:

1- The mean of the sampling distribution would be equal to the mean of the original distribution.

$$\mu_{\bar{x}} = \mu$$

2- The variance of the sampling distribution of the sample mean would be equal to the variance of the original distribution divided by n, where n is the size of the samples.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Click here for a CLT simulator.

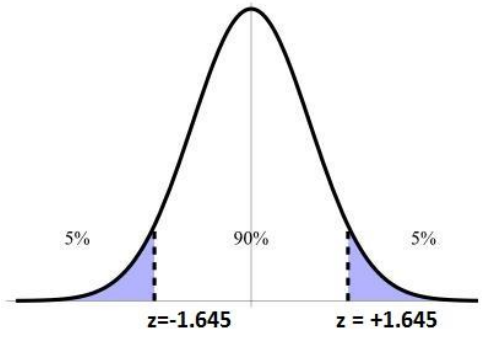
VIDEO TUTORIAL

Confidence Intervals Example

Suppose we randomly collected a sample of students with the following information:

- Sample size n = 50
- Sample mean weight x = 73.16 kg
- Population standard deviation s = 8.63

What is the 90% confidence interval for the true population mean weight?



- According to the z-table, the z-score for the 90% Confidence Interval = 1.64
- We find that value under the 0.9495 ~ 0.95 entry in the z table.
- The Margin of Error = 1.64 * (8.63/√50) = ~2
- Therefore, the 90% CI = 73.16 ± 2 = [71.16, 75.16]

z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608

Confidence Intervals Example

We interpret this 90% confidence interval as follows:

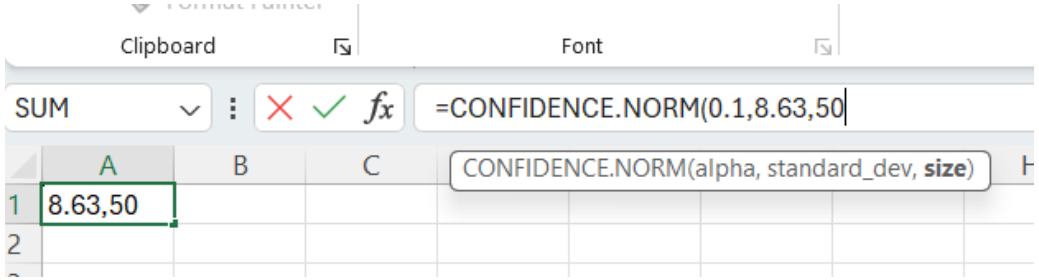
- The interval [71.16 kgs to 75.16 kgs] gives us a good estimate of where the mean lies, based on the collective information from our samples.
- This specific interval is our best estimate given the current data and the reliability of the method behind it.
- The 90% refers to the reliability of the method we are using, if we estimate the population parameter and computed the confidence intervals 100 times, we expect that 90% of these intervals will contain our parameter.

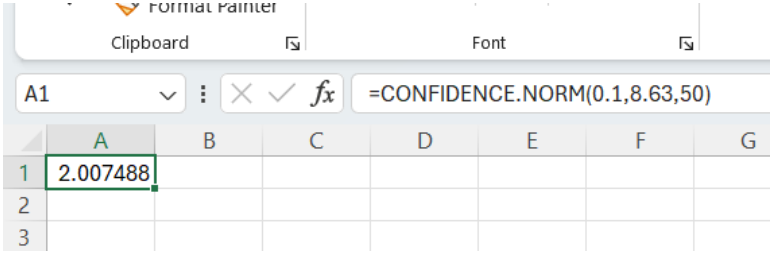
Solving Confidence Intervals using Excel Functions

Suppose we randomly collected a sample of students with the following information:

- Sample size n = 50
- Sample mean weight x = 73.16 kg
- Sample standard deviation s = 8.63

What is the 90% confidence interval for the true population mean weight?





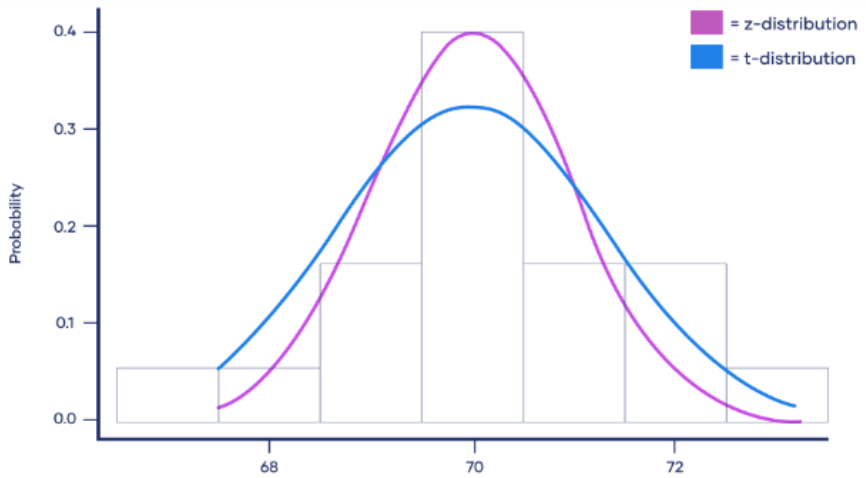
- We can compute the confidence interval using Excel CONFIDENCE.NORM function.
- This function computes the margin of error.
- CONFIDENCE.NORM(.1 , 8.63 , 50) = 2.0074 ~ 2
- Therefore, the 90% CI = 73.16 ± 2 = [71.16, 75.16]

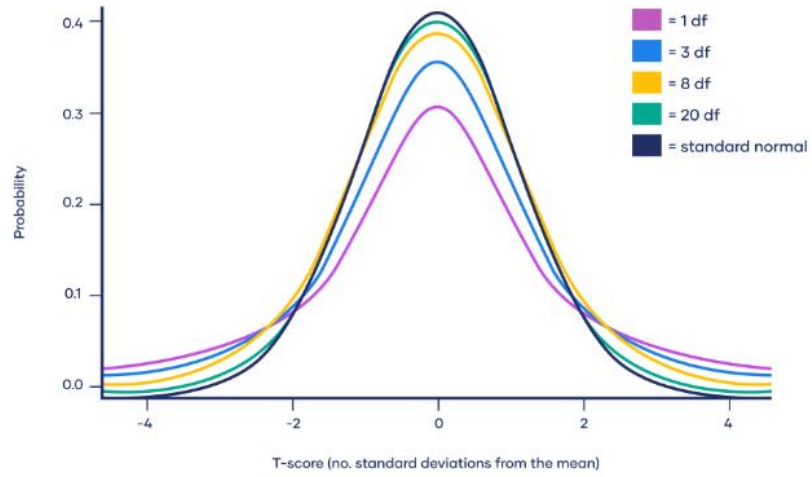
Transitioning to T-Distribution

- We can use the Z-distribution when the population variance is known or when our data is large enough ($n > 30$).
- However, real-world data often doesn't come with this information.
- In practice, the population variance is unknown, and we usually estimate the population variance using the sample variance, which introduces more variability into our statistics.
- The T-distribution accommodates this additional uncertainty and is especially useful for small sample sizes.
- The T-distribution often serves as a safer alternative because it accounts for the additional uncertainty from estimating the population variance.

Characteristics of T-Distribution

- The T-distribution, like the normal distribution, is bell-shaped and symmetric but has heavier tails.
- This means there is more probability in the tails and less in the center compared to a normal distribution.
- The exact shape of the T-distribution depends on the degrees of freedom (df), which are related to the sample size.
- Degrees of freedom in the context of the T-distribution refer to the number of independent values in a calculation of a statistic that are free to vary.
- For a confidence interval, df is equal to the sample size (n) minus 1 (df = n - 1).
- As the sample size increases, the T-distribution approaches the normal Z-distribution.





https://www.scribbr.com/statistics/t-distribution/

The T Table

- The first column, denoted "v," lists the degrees of freedom.
- Degrees of freedom typically equal the sample size minus one (n-1) and relate to the number of independent values in a set of observations.
- The top row lists different significance levels (α), like the probabilities in z-table.
- The intersection of a row and column gives the t-value or t score, which is the cutoff point on the t-distribution.
- The table lists the areas (probability) on the right side, of the t-scores (opposite to the z-tables), therefore, to find a t scores for confident intervals we split α by 2 and look out the resulting value in the table.
- For example, to find the 90% confidence interval for a sample of 20 observations, we find α which is 1 - .9 = .1
- We split α by 2 = .1 / 2 = 0.05 and we lookout that value in the table = 1.729

Table of the Student's *t*-distribution

The table gives the values of $t_{\alpha;\nu}$ where $\Pr(T_\nu > t_{\alpha;\nu}) = \alpha$, with ν degrees of freedom

ν \ α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Confidence Intervals Using T-Distribution

- The formula for a confidence interval using the T-distribution is like the one with the Z-distribution:

$$CI = \bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$$

- $\bar{x}$ is the sample mean
- s is the sample standard deviation
- n is the sample size
- $t_{\alpha/2}$ is the t-score from the T-distribution that corresponds to the desired confidence level.

Interval Estimates - Confidence Intervals

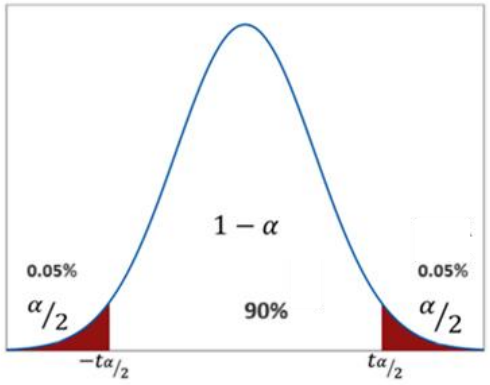
Suppose we randomly collected a sample of students with the following information:

- Sample size n = 20
- Sample mean weight x = 72.94 kg
- Samples standard deviation s = 8.3946

What is the 90% confidence interval for the true population mean weight?

- Confidence Interval 90% → *α = 1 - 0.9 = 0.1 → α/2 = 0.05*
- According to the t-table, the t-score for the 0.05 with df 20 – 1 = 19 → t score = 1.729
- The Margin of Error = 1.729 * (8.3946/√20) = 3.25
- Therefore, the 90% CI = 72.94 ± 3.25 = [69.69, 76.19]

ν \ α	0.1	0.05				
1	3.078	6.314				
2	1.886	2.920				
3	1.638	2.353				
4	1.533	2.132				
5	1.476	2.015				
6	1.440	1.943				
7	1.415	1.895				
8	1.397	1.860				
9	1.383	1.833				
10	1.372	1.812				
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.610
19	1.328	$\boxed{1.729}$	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450



What is the difference between the 90% Confidence Interval of the Z-Distribution and the 90% Confidence Interval of the T-Distribution?

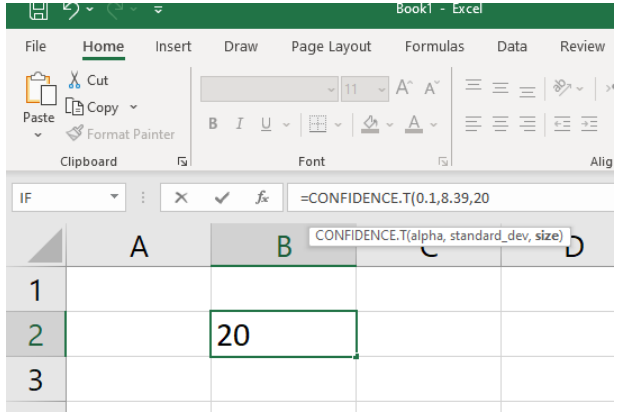
Solving Confidence Intervals using Excel Functions

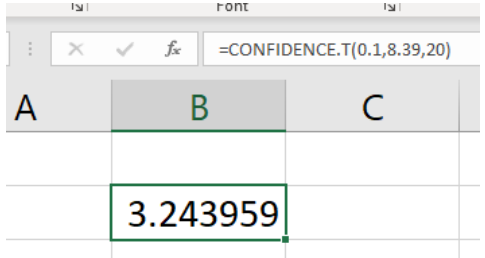
Suppose we randomly collected a sample of students with the following information:

- Sample size n = 20
- Sample mean weight x = 72.94 kg
- Samples standard deviation s = 8.3946

What is the 90% confidence interval for the true population mean weight?

- We can compute the confidence interval using Excel CONFIDENCE.T function.
- This function computes the margin of error.
- CONFIDENCE.T(.1 , 8. 39 , 20) = 3.24 ~ 2
- Therefore, the 90% CI = 72.94 ± 3.24 = [69.69, 76.19]





Hypothesis Testing

Hypothesis Testing

- A **hypothesis** is a claim about a population parameter that can be examined.
- Hypothesis Testing is a statistical method that uses sample data to evaluate a hypothesis about a population parameter.
- The purpose of hypothesis testing is to determine whether there is enough evidence in a sample of data to support a particular belief about a population.
- Examples of claims that can be checked:
  - The height of students in the university is equal to 170 cm.
  - The average score for political science students is less than the average score of social science students.
  - The proportion of female students is larger than the proportion of male students.

The Null and The Alternative Hypothesis

- When formulating a hypothesis test, we define two hypothesis, the null hypothesis (H_0) and the alternative hypothesis (H_1 or H_a).
- These two hypothesis complements each other, only one of them can be valid.
- For example, to test our claim that the average height of university students is 170 cm, we define the two hypothesis below, knowing that only one of can be valid, either the first or the second but not both:
  - *The Null hypothesis* H_0*: The average height of students IS EQUAL TO* $170cm.$
  - *The Alternative hypothesis* H_a*: The average height of students IS NOT* $170cm.$
- Hypothesis are often expressed with symbols like this:

$$H_0: \mu = 170\ cm$$

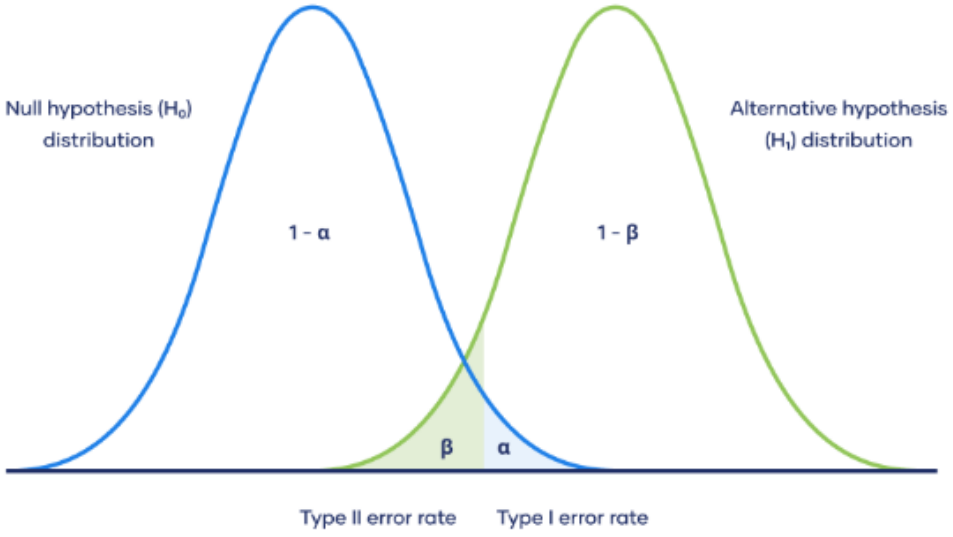
$$H_a: \mu \neq 170\ cm$$

- Notice that the null hypothesis is framed in a way to create a baseline that there is no difference or no effect, or in this context, that the mean height is not higher than $170\ cm$.
- Then, the burden of proof is on the alternative hypothesis to demonstrate with sufficient evidence that the mean age is indeed higher.

Type I and Type II Errors

Type I and Type II errors are two potential forms of incorrect conclusions in hypothesis testing.

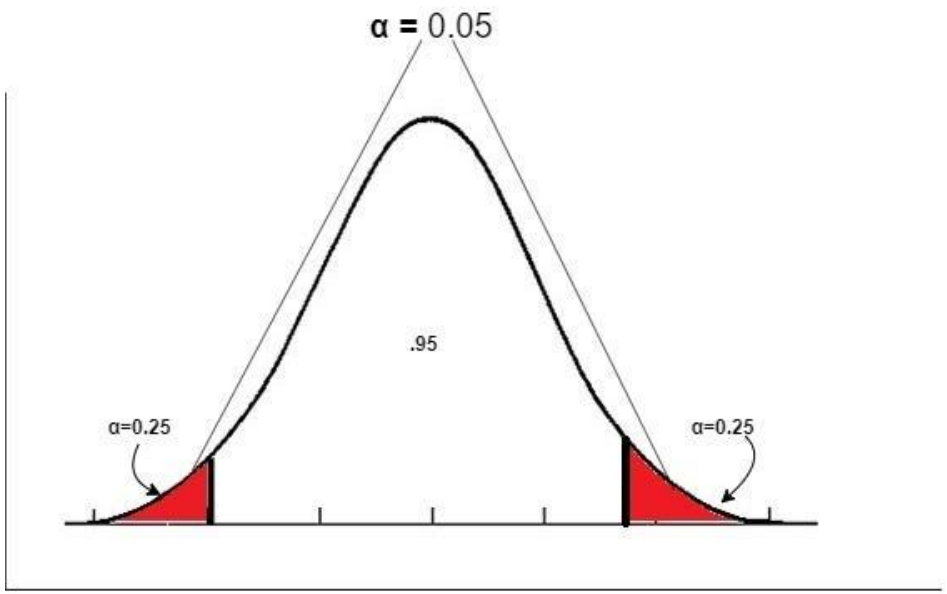
- Type I Error (False Positive): Rejecting H_0 when it is actually true.
- Type II Error (False Negative): Failing to reject H_0 when it is false.
- The acceptable level of Type I error is designated by alpha (α), while the acceptable level of Type II error is designated beta (β).



- A Type I error occurs when the null hypothesis is incorrectly rejected when it is actually true.
- Conversely, a Type II error happens when the null hypothesis is not rejected when it is in fact false, akin to a false negative, where we fail to detect a true effect or difference.
- Consider a pharmaceutical company deciding whether a new drug is effective. A Type I error would occur if the company concludes the drug works when it does not, potentially leading to a costly recall and reputation damage when the truth emerges.
- On the other hand, a Type II error would be failing to recognize the drug's efficacy, missing out on the opportunity to bring a beneficial product to market, which can result in lost revenue and competitive disadvantage. The company must balance these risks, often prioritizing the avoidance of one type of error over the other based on the potential costs and impacts associated with each.

The Significance Level α

- Before conducting the test, we need to set the significance level α.
- The significance level is a measure of the strength of the evidence that must be present in our sample to reject the null hypothesis and conclude that the effect is statistically significant.
- Typical significance levels are: α=0.1 (10%), α=0.05 (5%), α=0.01 (1%)
- A lower significance level means that the evidence in the data needs to be stronger to reject the null hypothesis.



There is no "correct" significance level - it only states the amount of uncertainty of the conclusion.

A 5% significance level means that we expect to reject a true null hypothesis 5 out of 100 times (Type I Error).

The Test Statistic

- The test statistic is the value we compute based on the sample data that we collect.
- The test statistic is used to decide the outcome of the hypothesis test.
- We standardize test statistic using:
  - Z Standard Normal Distribution for large samples or when the variance is known.
  - The Student's T-Distribution (T) for small samples and the population variance is unknown.

Computing the Test Statistic

Populations	Variance	Samples	Statistic	Variance	Statistic
One	Known	-	Z	σ^2	$Z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$
One	Unknown	-	t	s^2	$t = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}}$
Two	Known	Independent	Z	σ_x^2, σ_y^2	$Z = \frac{(\bar{x}-\bar{y})-\mu_0}{\sqrt{\frac{\sigma_x^2}{n_x}+\frac{\sigma_y^2}{n_y}}}$
Two	Unknown, Assumed Equal	Independent	t	$s_p^2 = \frac{(n_x-1)s_x^2+(n_y-1)s_y^2}{n_x+n_y-2}$	$t = \frac{(\bar{x}-\bar{y})-\mu_0}{\sqrt{s_p^2\left(\frac{1}{n_x}+\frac{1}{n_y}\right)}}$
Two	Unknown, Not Assumed Equal	Independent	t	s_x^2, s_y^2	$t' = \frac{(\bar{x}-\bar{y})-\mu_0}{\sqrt{\frac{s_x^2}{n_x}+\frac{s_y^2}{n_y}}}$
Two	Unknown	Dependent/Paired	t	s_d^2	$t = \frac{\bar{d}-\mu_0}{\frac{s_d}{\sqrt{n}}}$

The Critical Value and the p-Value Approach to Hypothesis Testing

- To decide whether to reject the null hypothesis, a test statistic is calculated.
- The decision is made based on the numerical value of that test statistic.
- There are two approaches how to arrive at that decision: The critical value approach and the p-value approach.

The critical value approach

- By applying the critical value approach, it is determined, whether the observed test statistic is more extreme than a defined critical value.
- Therefore, the observed test statistic (calculated based on sample data) is compared to the critical value (a kind of cutoff value).
- If the test statistic is more extreme than the critical value, the null hypothesis is rejected.
- If the test statistic is not as extreme as the critical value, the null hypothesis is not rejected.
- The critical value is computed based on the given significance level α and the type of probability distribution of the idealized model.
- The critical value divides the area under the probability distribution curve in rejection region(s) and non-rejection region.

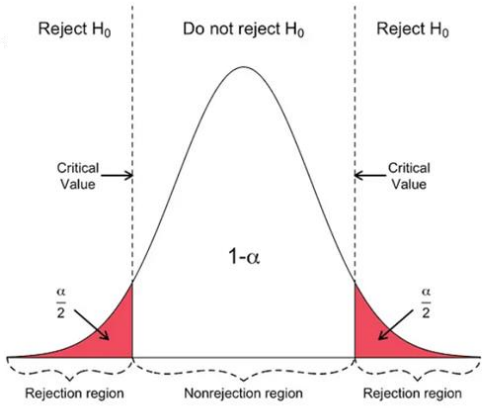
References

The Critical Value Approach to Hypothesis Testing

- The following three figures show a two tailed test, a left tailed test and a right sided test.
- In a two-sided test the null hypothesis is rejected if the test statistic is either too small or too large. Thus, the rejection region for such a test consists of two parts: one on the left and one on the right.
- For a left-tailed test the null hypothesis is rejected, if the test statistic is too small. Thus, the rejection region for such a test consists of one part, which is left from the center.
- For a right-tailed test the null hypothesis is rejected, if the test statistic is too large. Thus, the rejection region for such a test consists of one part, which is right from the center.
- Note: if p≤α , reject H_0 ; otherwise, if p>α , do not reject H_0

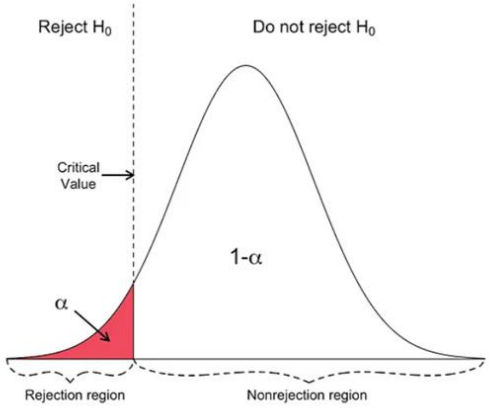
Two Tails Test

$$H_0 : \mu_X = \mu_0$$
$$H_1 : \mu_X \neq \mu_0$$



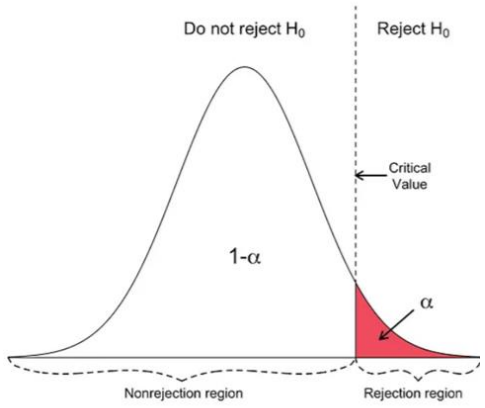
One Tail Test - Left Tail

$$H_0 : \mu_X = \mu_0$$
$$H_1 : \mu_X < \mu_0$$



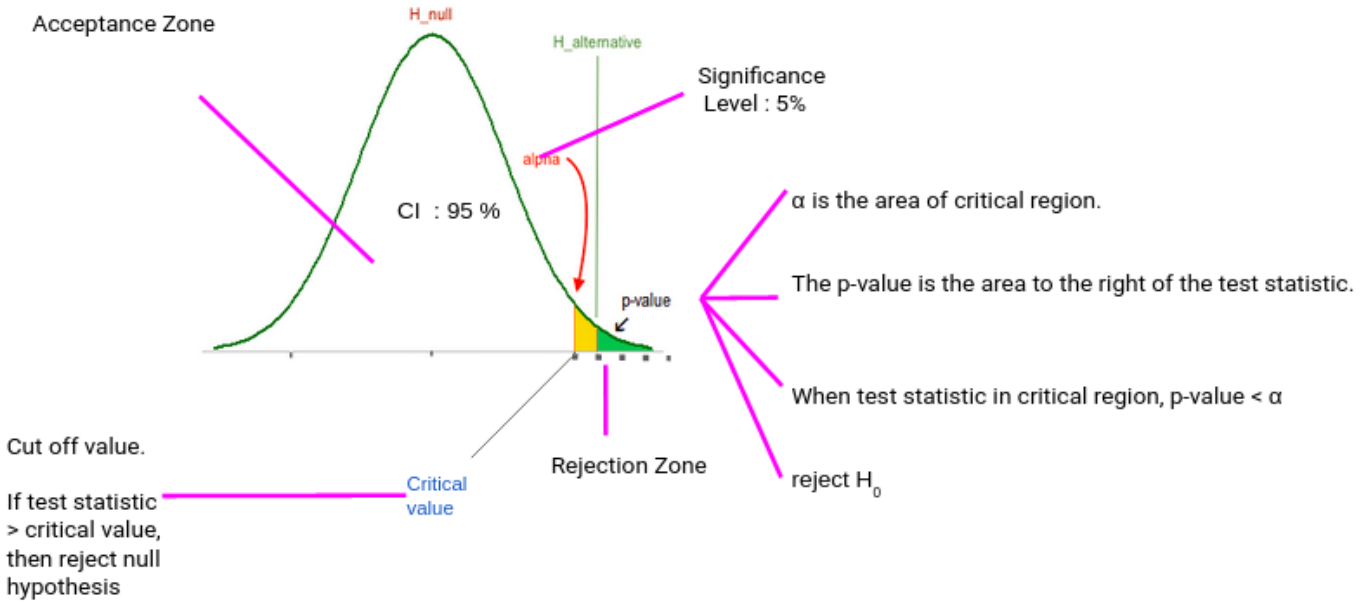
One Tail Test - Right Tail

$$H_0 : \mu_X = \mu_0$$
$$H_1 : \mu_X > \mu_0$$



The p-Value Approach to Hypothesis Testing

- For the p-value approach the likelihood (p-value – green region) of the numerical value of the test statistic is compared to the specified significance level (α – yellow region) of the hypothesis test.
- The p-value corresponds to the probability of observing sample data at least as extreme as the obtained test statistic.
- Small p-values provide evidence against the null hypothesis.
- The smaller (closer to 0) the p-value, the stronger is the evidence against the null hypothesis.
- If the p-value is less than or equal to the specified significance level α , the null hypothesis is rejected. Otherwise, the null hypothesis is not rejected.



References

Statistical Testing Steps

1. State the hypotheses: Define H0 and H1.
2. Set the significance level: Choose α (commonly 0.05).
3. Collect data: Obtain and prepare your data set.
4. Perform the test: Calculate the test statistic and P-value.
5. Make a decision: Reject H0 if the P-value is less than α; otherwise, we fail to reject H0.

Note: We cannot we accept the null hypothesis, we can only fail to reject it. The basis for this testing process comes from logic. For example, if our null hypothesis = All students live in Amman, it will ne challenging to check if all students live in Amman, it will be much easier to find just one student who does not live in Amman and therefore we can safely reject the null hypothesis. But if our sample students also lived in Amman, we cannot accept the null hypothesis because there are many students we couldn't check, so we just fail to reject the null hypothesis.

Types of Hypothesis Tests

- Z-test: Used for normally distributed data when the variance is known.
  - One-sample z-test: Tests the mean of a single group against a known mean.
  - Two-sample z-test: Compares the means of two dependent groups (paired test).
  - Two-sample z-test: Compares the means of two independent groups.
- T-test: Used for normally distributed data when the variance is unknown.
  - One-sample t-test: Tests the mean of a single group against a known mean.
  - Two-sample t-test: Compares the means of two dependent groups (paired test).
  - Two-sample t-test: Compares the means of two independent groups (variance assumed to be equal).
  - Two-sample t-test: Compares the means of two independent groups (variance assumed to be not equal).
- ANOVA (Analysis of Variance): Used to compare the means of three or more samples.
- Chi-square test: Tests for relationships between categorical variables.

Hypothesis Testing for One Population Mean

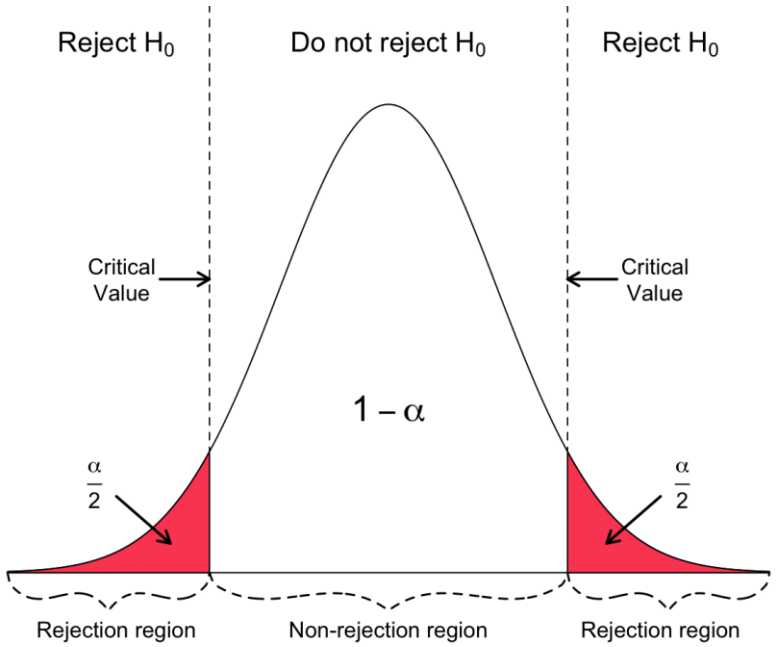
One Sample - Two Tailed Test

- Suppose that we have a general believe that the average students age is 21.
- Someone is claiming that the average age <> 21.
- To test that claim, we define the null and the alternative hypotheses as follows:

$$H_0: \mu = 21$$

$$H_a: \mu <> 21$$

- We collect data, and we compute the test statistic.
- If the test statistic is within the rejection region, we reject the null hypothesis, otherwise we fail to reject the null hypothesis.



References

One Sample - Two Tailed Test

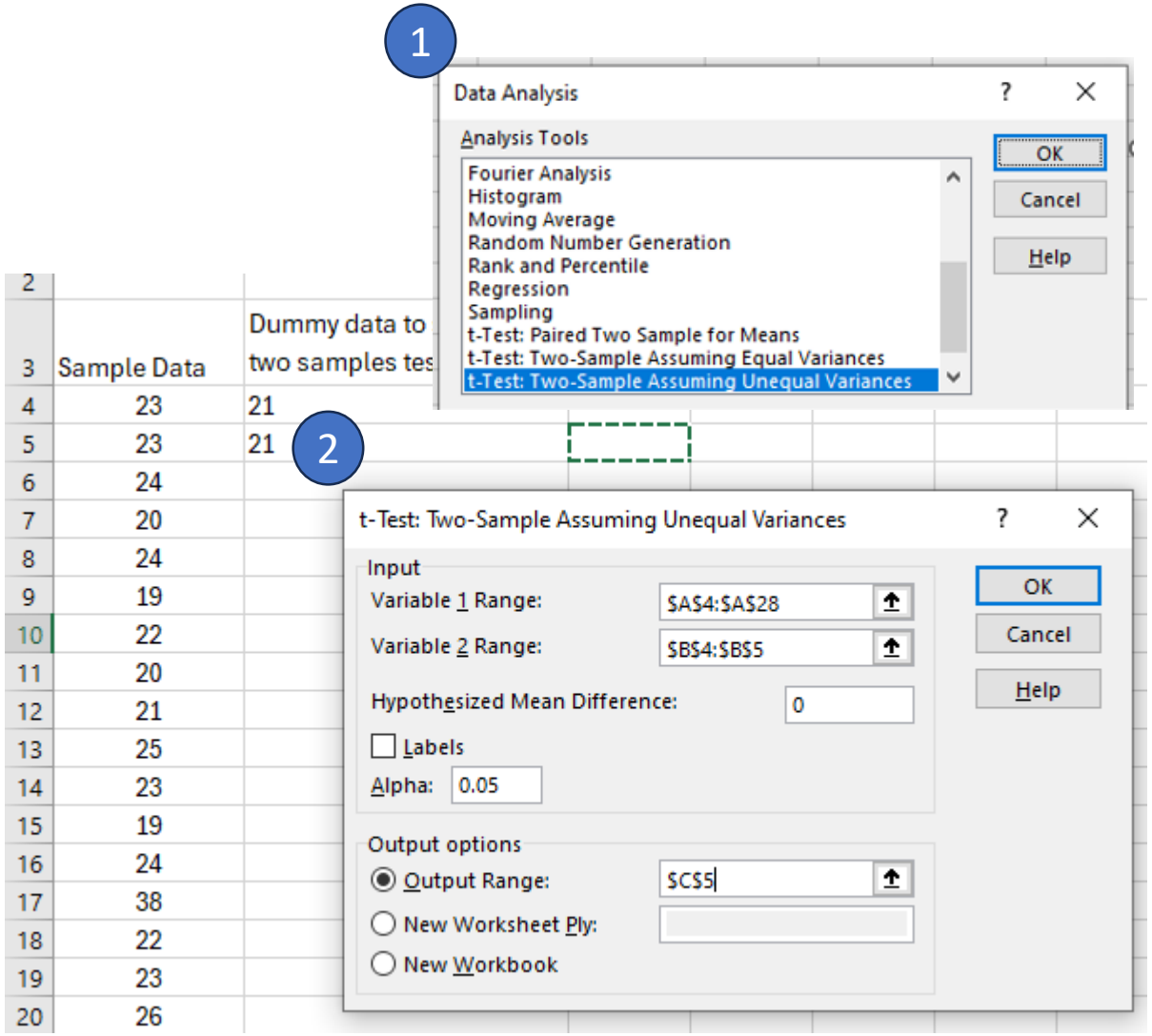
Is there enough evidence (at α = .05) to conclude that the population mean age for the university students is different than 21 based on the following sample data?

23	19	23	23	21
23	22	19	26	23
24	20	24	20	26
20	21	38	19	20
24	25	22	21	23

To test whether average students' age is equal to 21, we define the null and alternative hypotheses as follows:

$$H_0: \mu = 21$$

$$H_a: \mu <> 21$$



One Sample - Two Tailed Test

- According to the test results, we reject the null hypothesis that the mean is not equal to 21.
- Using our sample data, we found that the average students age is more than 21 with significant evidence.
- We reached this conclusion because the t statistic is larger than the critical value (falls inside the rejection region).
- We also reached that conclusion because the p-value (0.0294) which is less than 0.05.

	Age	Dummy
Mean	22.76	21
Variance	14.44	0
Observations	25	2
Hypothesized Mean Difference	0	
df	24	
t Stat	**2.315789474**	
P(T<=t) one-tail	0.014715837	
t Critical one-tail	1.71088208	
P(T<=t) two-tail	**0.029431673**	
t Critical two-tail	2.063898562	

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

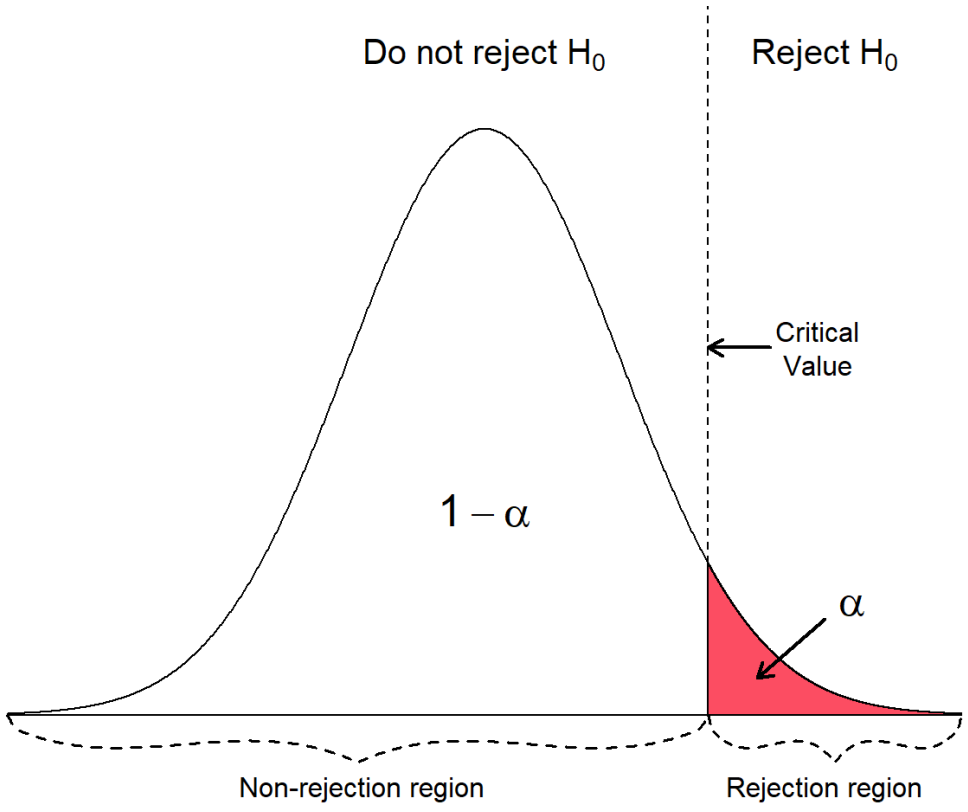
One Sample - One Tailed Test (Upper Tail Test)

- Suppose that we have a general believe that the average students age is 21.
- Someone is claiming that the average age > 21.
- To test that claim, we define the null and the alternative hypotheses as follows:

$$H_0: \mu \leq 21$$

$$H_a: \mu > 21$$

- We collect data, and we compute the test statistic.
- If the test statistic is within the rejection region, we reject the null hypothesis, otherwise we fail to reject the null hypothesis.



References

One Sample - One Tailed Test (Upper Tail Test)

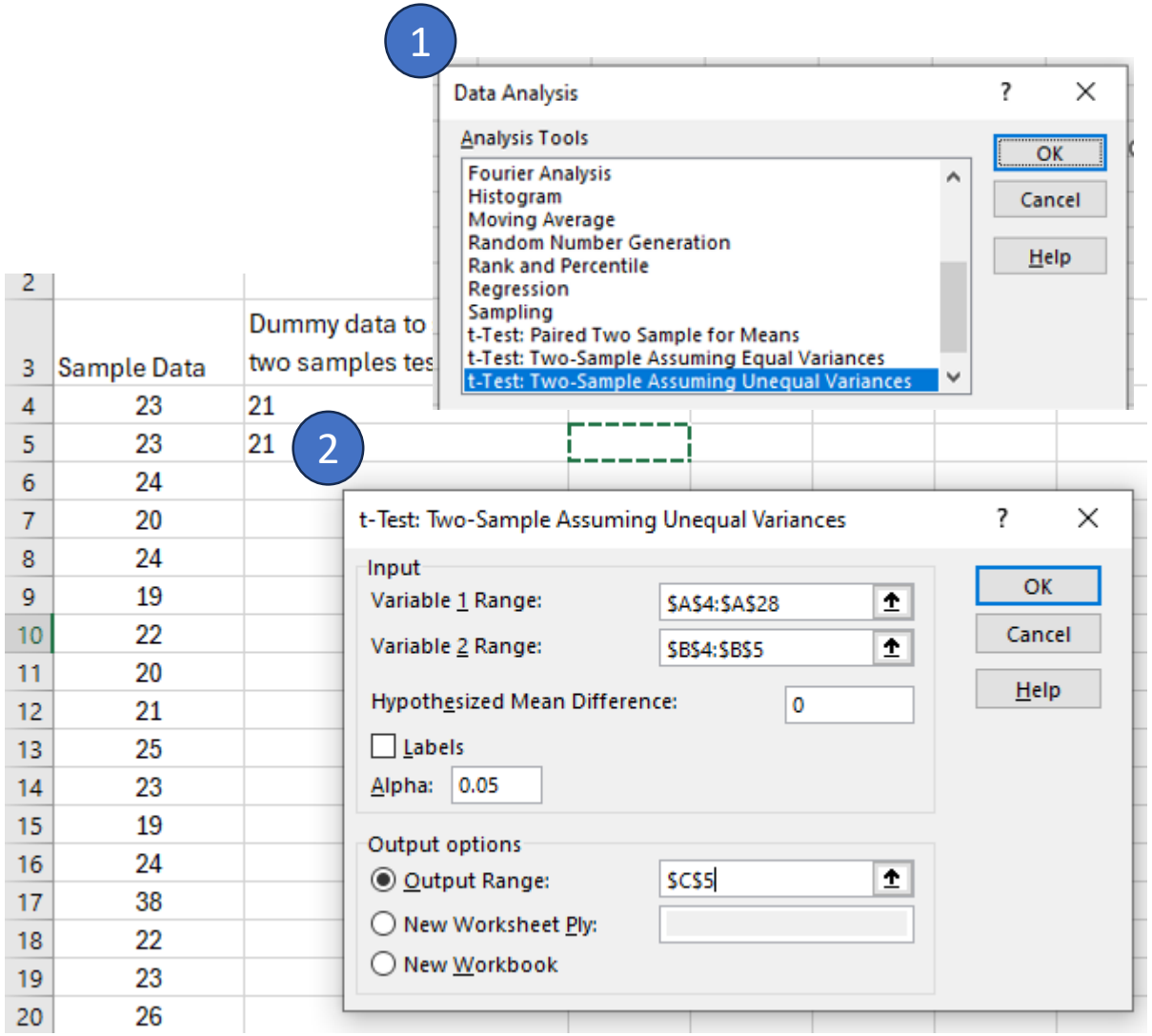
Is there enough evidence (at α = .05) to conclude that the population mean age for the university students is higher than 21 based on the following sample data?

23	19	23	23	21
23	22	19	26	23
24	20	24	20	26
20	21	38	19	20
24	25	22	21	23

To test whether average students' age is more than 21, we define the null and alternative hypotheses as follows:

$$H_0: \mu \leq 21$$

$$H_a: \mu > 21$$



One Sample - One Tailed Test (Upper Tail Test)

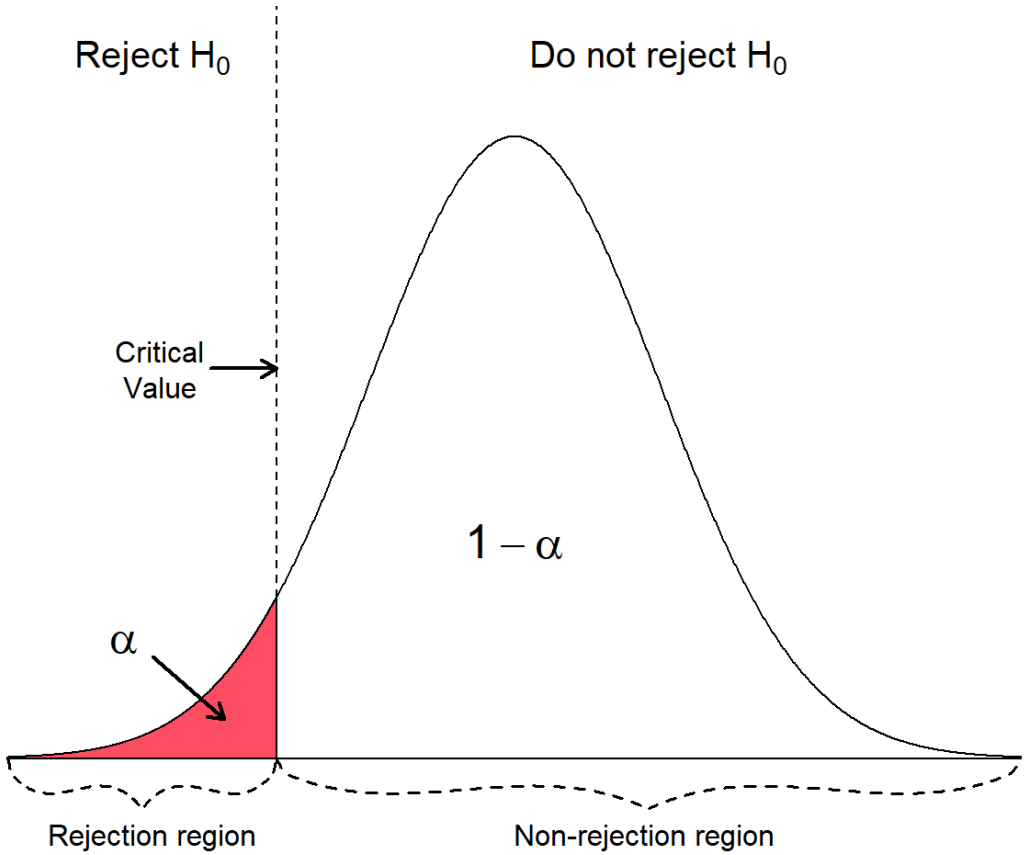
- According to the test results, we reject the null hypothesis that the mean is less than or equal to 21.
- Using our sample data, we found that the average students age is more than 21 with significant evidence.
- We reached this conclusion because the t statistic is larger than the critical value (falls inside the rejection region).
- We also reached that conclusion because the p-value (0.014) which is less than 0.05.

	Age	Dummy
Mean	22.76	21
Variance	14.44	0
Observations	25	2
Hypothesized Mean Difference	0	
df	24	
t Stat	**2.315789474**	
P(T<=t) one-tail	**0.014715837**	
t Critical one-tail	**1.71088208**	
P(T<=t) two-tail	0.029431673	
t Critical two-tail	2.063898562	

One Sample - One Tailed Test (Lower Tail Test)

Is there enough evidence (at α = .05) to conclude that the population mean height for the university students is less than 170 based on the following sample data?

175.65	164.61	165.96	168.59
171.82	167.32	175.88	167.56
166.08	167.17	164.32	163.92
167.49	175.20	172.61	176.43
169.61	163.77	171.58	174.42



To test whether average students' height is less than 170 cm, we define the null and alternative hypotheses as follows:

$$H_0: \mu \geq 170\ cm$$

$$H_a: \mu < 170\ cm$$

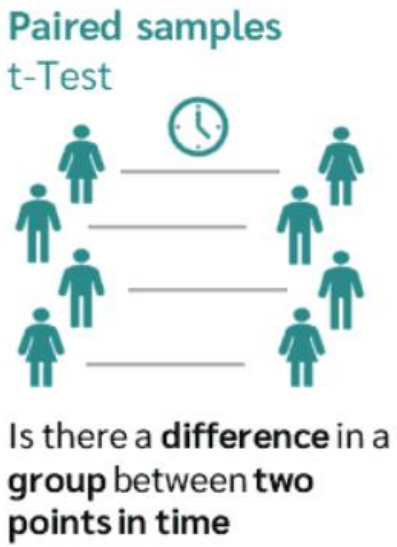
Hypothesis Testing for Two Populations Means

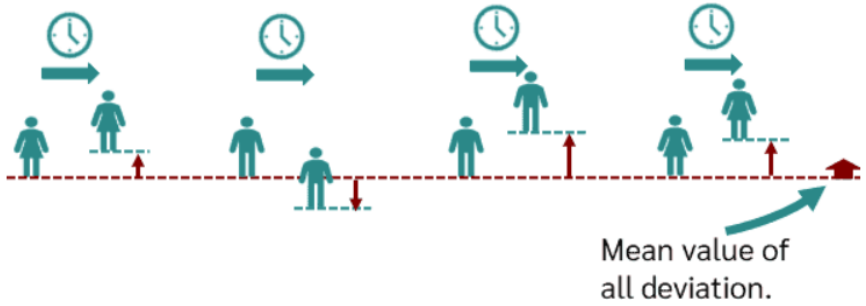
Types of Hypothesis Tests

- Z-test: Used for normally distributed data when the variance is known.
  - One-sample z-test: Tests the mean of a single group against a known mean.
  - Two-sample z-test: Compares the means of two dependent groups (paired test).
  - Two-sample z-test: Compares the means of two independent groups.
- T-test: Used for normally distributed data when the variance is unknown.
  - One-sample t-test: Tests the mean of a single group against a known mean.
  - Two-sample t-test: Compares the means of two dependent groups (Paired t test).
  - Two-sample t-test: Compares the means of two independent groups (variance assumed to be equal).
  - Two-sample t-test: Compares the means of two independent groups (variance assumed to be not equal).
- ANOVA (Analysis of Variance): Used to compare the means of three or more samples.
- Chi-square test: Tests for relationships between categorical variables.

Compares the means of two dependent groups (Paired T Test)

- The dependent samples T-test (or paired samples t-test) is a statistical test that determines whether there is a difference between two dependent groups or samples.
- We need the paired t-test whenever we survey the same group or sample at two points in time.
- This can be the case, for example, in longitudinal studies with several measurement points (time series analyses) or in intervention studies with experimental designs (before-after measurement).
- An example of dependent sampling is when the weight of a group of people is measured at two points in time (before and after going into a diet).
- The weight of a person can be measured individually before and after going into the diet, and the difference in the measured values can be calculated in each case.





References

Compares the means of two dependent groups (Paired T Test)

- We are interested in the question whether an online statistics learning tutorial helps students improve their grades.
- There are three variables of interest in the student's data set (score1, score2 and online.tutorial).
- The variables score1 and score2 show the grades (0-100) for two exams on mathematics and statistics. The higher the value the better the student performed.
- The variable online.tutorial is a binary variable, which is 1 if the student completed the online statistics learning tutorial, or 0 otherwise.
- The first exam takes place before the students attended the online statistics learning tutorial.
- The participation in the online statistics learning tutorial is not mandatory, however the two exams are obligatory for all students.
- We want to examine if the group of students, which attended the online statistics learning tutorial, performs better on the second exam compared to the first exam.

Random sample of students who took the online training

`	Score 1	Score 2
1	67	71
2	62	64
3	60	57
4	53	52
5	75	77
6	64	79
7	43	48
8	74	70
9	59	53
10	54	58
11	66	66
12	81	90
13	55	57
14	64	68
15	72	71
16	76	78
17	75	73
18	90	92
19	83	90
20	64	73
21	49	51
22	41	51
23	75	87
24	66	66
25	75	74
26	49	57
27	91	89
28	68	69
29	72	72
30	87	93

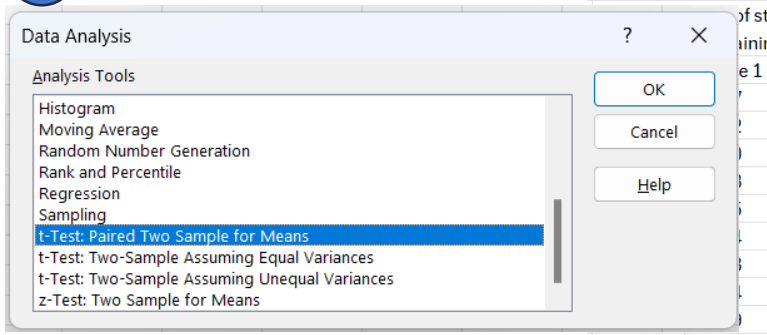
$H_0: \mu_{score2} \leq \mu_{score1}$ *Null Hypothesis states that there is no effect for the intervention*

$H_1: \mu_{score2} > \mu_{score1}$ *Alternative Hypothesis claims that the intervention caused an effect*

References

Compares the means of two dependent groups (Paired T Test)

1



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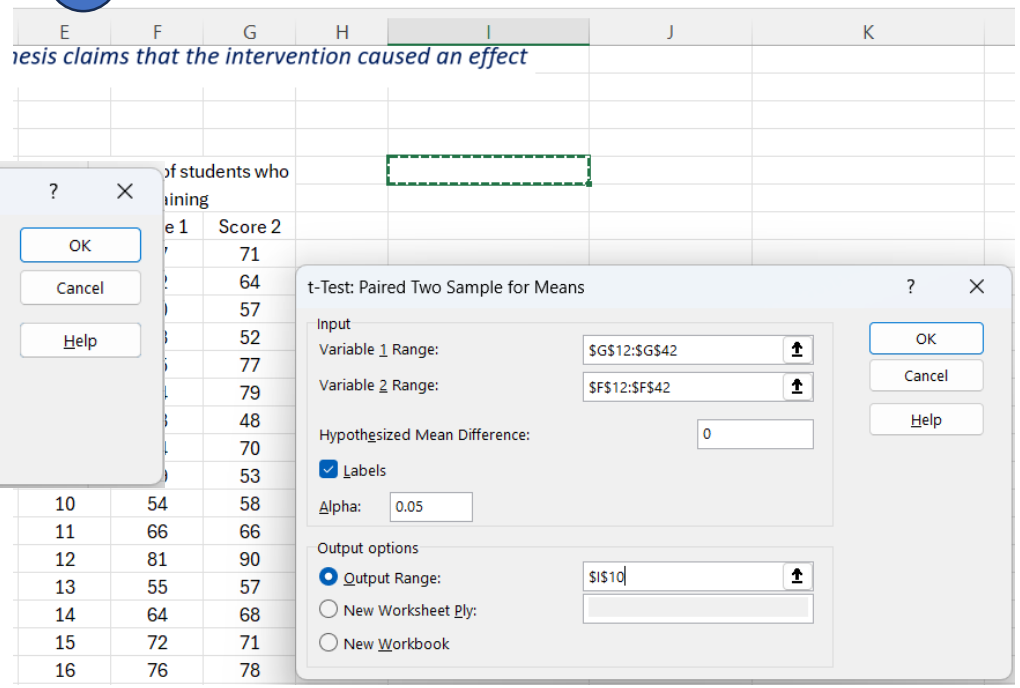


3

t-Test: Paired Two Sample for Means

	Score 2	*Score 1*
Mean	69.86666667	67
Variance	180.6712644	171.3793103
Observations	30	30
Pearson Correlation	0.931421115	
Hypothesized Mean Diff	0	
df	29	
t Stat	**3.187976131**	
P(T<=t) one-tail	**0.00171094**	
t Critical one-tail	**1.699127027**	
P(T<=t) two-tail	0.003421879	
t Critical two-tail	2.045229642	

- Based on the sample data and the test statistic, we found that there is a statistically significant difference between the before and after score, the after score (score2) is higher than score1.
- This finding is supported by the test statistic and the p-value, therefore we REJECT the null hypothesis, and we conclude that the online tutorial helped students in enhancing their exam performance.

References

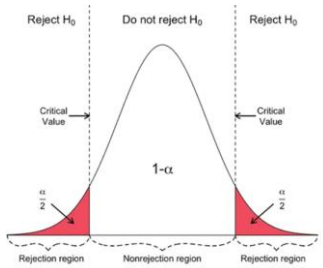
Compares the means of two independent groups (variance assumed to be equal)

- In this section we perform a hypothesis test for the means of two populations.
- We examine the mean annual salary (in Euro) of graduates.
- The first population consist of male students and the second population of female students.
- The standard deviations of the two populations are unknown but assumed to be equal.
- The question is, whether there is a difference (at α 5%) in the mean annual salary of graduates related to gender (two tailed test)

	Male Random Sample	Female Random Sample
1	$39,647.68	$29,543.60
2	$55,726.10	$30,972.76
3	$34,717.77	$33,216.12
4	$29,999.35	$24,143.27
5	$50,763.66	$40,466.69
6	$52,433.73	$29,543.60
7	$44,385.84	$47,760.48
8	$36,599.27	$45,054.95
9	$47,842.44	$42,742.79
10	$30,854.42	$42,202.39
11	$58,614.72	$40,606.41
12	$62,887.03	$29,783.76
13	$48,288.53	$46,945.17
14	$43,191.15	$29,555.94
15	$43,756.58	$45,042.30
16	$48,453.00	$29,555.94
17	$23,090.67	$45,625.09
18	$55,660.20	$28,017.87
19	$57,527.17	$35,840.90
20	$65,776.23	$33,648.38
21	$50,011.69	$46,858.67
22	$46,213.64	$42,463.48
23	$42,211.41	$33,200.27
24	$61,384.55	$28,338.17
25	$49,780.18	$26,265.35
26	$50,474.57	$34,373.99
27	$43,281.92	$37,312.85
28	$55,484.64	$37,709.73
29	$24,409.73	$35,106.01
30	$40,489.49	$37,807.11

$H_0: \mu_{male} = \mu_{female}$ *Null Hypothesis states that there is no difference*

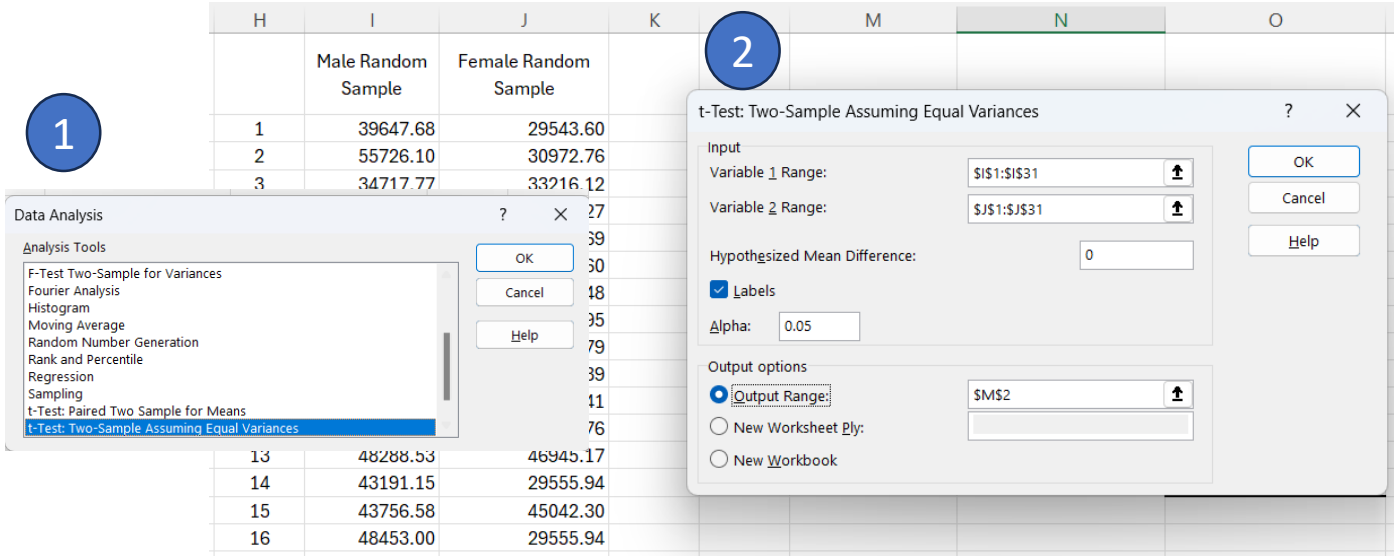
$H_1: \mu_{male} \neq \mu_{female}$ *Alternative Hypothesis claims that there is a difference*



References

Compares the means of two independent groups (variance assumed to be equal)

1



3

t-Test: Two-Sample Assuming Equal Variances

	Male Random Sample	*Female Random Sample*
Mean	46465.24594	36323.46873
Variance	117433271.4	48853686.8
Observations	30	30
Pooled Variance	83143479.09	
Hypothesized Mean	0	
df	58	
t Stat	**4.30770147**	
P(T<=t) one-tail	3.22648E-05	
t Critical one-tail	1.671552762	
P(T<=t) two-tail	**6.45296E-05**	
t Critical two-tail	**2.001717484**	

- Based on the sample data and the test statistic, we found that there is a statistically significant difference between male salaries and female salaries.
- This finding is supported by the test statistic and the p-value, therefore we REJECT the null hypothesis, and we conclude that the salaries of male graduates and female graduates are not equal.

References

Compares the means of two independent groups (variance assumed to be not equal)

- In this section we perform a hypothesis test for the means of two populations.
- To showcase the non-pooled t-test we test a claim that states that the annual salaries for political science female students is higher than the annual salaries of the social science female students.
- The standard deviations of the two populations are unknown and assumed to be unequal.
- We want to test at α 1%, whether the mean political science salary is higher than the mean salary of the social science (one tail test, upper tail).

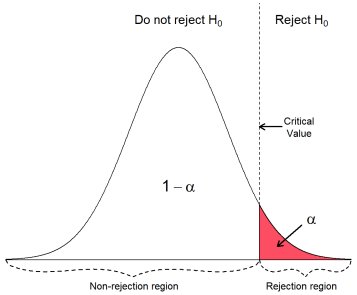
	Political Science Random Sample	Social Science Random Sample
1	$30,796.16	$33,641.73
2	$32,038.41	$29,635.67
3	$32,038.41	$38,738.01
4	$32,996.39	$21,453.99
5	$33,648.38	$31,484.80
6	$41,430.16	$34,287.96
7	$40,032.78	$35,583.43
8	$39,710.80	$28,672.85
9	$36,911.26	$30,062.71
10	$45,900.13	$34,420.48
11	$35,270.19	$33,641.73
12	$43,318.41	$26,291.32
13	$41,282.13	$31,151.42
14	$41,430.16	$38,738.01
15	$21,485.62	$33,565.74
16	$29,195.50	$31,484.80
17	$45,141.81	$32,194.59
18	$34,492.08	$35,746.22
19	$27,893.26	$28,281.25
20	$38,591.54	$31,405.48
21	$33,726.58	$32,194.59
22	$29,832.22	$35,987.41
23	$32,154.94	$35,583.43
24	$34,686.49	$34,227.49
25	$25,607.82	$25,758.88
26	$33,726.58	$34,373.99
27	$37,782.86	$25,658.83
28	$35,507.83	$34,201.58
29	$33,955.21	$31,151.42
30	$25,883.03	$33,641.73

$H_0: \mu_{political\ science} \leq \mu_{social\ science}$

Null Hypothesis states that the salaries are equal

$H_1: \mu_{political\ science} > \mu_{social\ science}$

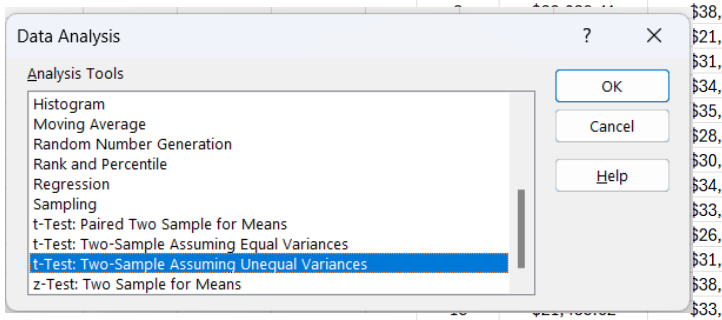
Alternative Hypothesis claims that political science is higher



References

Compares the means of two independent groups (variance assumed to be not equal)

1



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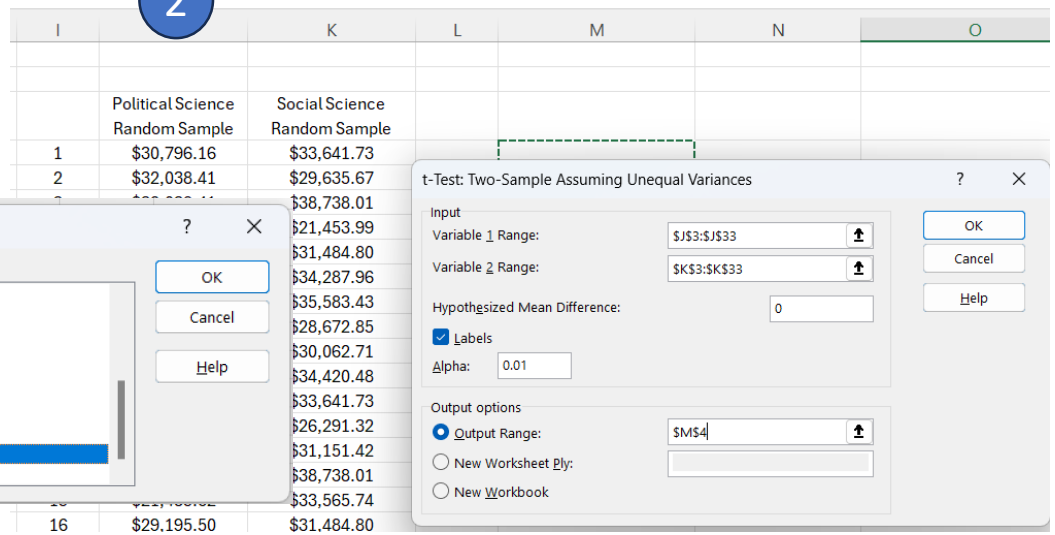


3

t-Test: Two-Sample Assuming Unequal Variances

	Political Science Random Sa	*Social Science Random Sample*
Mean	34882.23824	32108.71836
Variance	34602476.07	15267236.06
Observations	30	30
Hypothesized Mean Differe	0	
df	50	
t Stat	**2.151163793**	
P(T<=t) one-tail	**0.018159982**	
t Critical one-tail	**2.403271917**	
P(T<=t) two-tail	0.036319964	
t Critical two-tail	2.677793271	

- Based on the sample data and the test statistic, we found that there is no statistically significant difference (at α 1%) between the salaries of political science and social science.
- This finding is supported by the test statistic and the p-value, therefore we FAIL TO REJECT the null hypothesis which states that the salaries of political science and social science are equal for female graduates.

References

Hypothesis Testing for More than Two Populations Means

One Way ANOVA

Introduction to One-Way ANOVA

- Definition: One-Way ANOVA is a statistical test that compares the means of three or more independent groups to determine if at least one group mean is statistically different from the others.
- Purpose: To test for significant differences between the means of multiple groups when you have one categorical independent variable and one continuous dependent variable.
- Appropriate Use Cases:
  - Comparing customer satisfaction levels across different store locations.
  - Examining student performance across different teaching methods.
- Key Advantage:
  - Allows simultaneous comparison of more than two groups, providing a broader analysis than t-tests.

ANOVA

Hypothesis in One-Way ANOVA:

- Null Hypothesis (H0): All group means are equal.
- Alternative Hypothesis (H1): At least one group mean is different from the others.

How to Calculate?

1. Calculate Group Means and Overall Mean
2. Compute Sum of Squares:
   - Total Sum of Squares (SST): Variability of all data points around the overall mean.
   - Within-Group Sum of Squares (SSW): Variability within each group.
   - Between-Group Sum of Squares (SSB): Variability due to the interaction between the groups.

- Calculate Mean Squares:
  - Mean Square Between (MSB): SSB divided by the degrees of freedom between groups (k-1).
  - Mean Square Within (MSW): SSW divided by the degrees of freedom within groups (N-k).
- F-Statistic:
  - F = MSB / MSW

ANOVA Example

- Scenario: A school tests three different teaching methods to determine which one results in the best performance on a standardized test.
- The students are randomly assigned to one of three groups, each receiving instruction through a different method. Their scores on a standardized test are recorded.
- Variables:
  - Group: The teaching method group (A, B, C)
  - Score: The score each student received on the standardized test.

Student ID	Group	Score
1	A	82
2	A	88
3	A	84
4	A	90
5	A	85
6	B	78
7	B	75
8	B	80
9	B	82
10	B	77
11	C	90
12	C	92
13	C	95
14	C	91
15	C	94

Interpreting Results

- F-Statistic and P-Value:
  - An F-Statistic higher than the critical value or a P-value less than the alpha level (e.g., 0.05) indicates significant differences between group means.
- Post Hoc Tests:
  - If the ANOVA is significant, perform post hoc tests (e.g., Tukey's HSD) to determine which specific groups differ from each other.

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Group A	5	429	85.8	10.2
Group B	5	392	78.4	7.3
Group C	5	462	92.4	4.3

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	490.5333	2	245.2667	33.75229	1.18232E-05	3.885293835
Within Groups	87.2	12	7.266667			
Total	577.7333	14				

Both the F-Statistic and the P-value indicate significant difference between groups means, therefore we reject the null hypothesis.

Hypothesis Testing for Categorical Data

Chi-Square Tests

Introduction to Chi-Square Tests

- Definition: Chi-Square tests are statistical methods used to determine if there is a significant association between categorical variables.
- Types of Chi-Square Tests:
- Test of Independence: Determines if two or more categories are independent.
- Goodness of Fit Test: Determines if a sample matches an expected distribution.

Focus on Chi-Square Test of Independence

- Purpose: To assess whether observed frequencies in a contingency table differ significantly from expected frequencies derived from hypothesis.
- Applicable Scenarios: Useful in marketing research, opinion polling, healthcare, and demographic studies.

Assumptions of the Chi-Square Test of Independence

- Sample Adequacy: The sample size should be large enough, typically each category should have an expected frequency of 5 or more.
- Independence: Observations must be independent of each other.
- Random Sampling: Data should be collected randomly.
- Data Level: The test requires nominal or ordinal data.

The Chi-Square Statistic

- **Formula:** $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$
  - O_i: Observed frequency
  - E_i: Expected frequency
- **Degrees of Freedom:** $(r-1) \times (c-1)$
  - r: Number of rows in the table
  - c: Number of columns in the table

Calculating Expected Frequencies

- **Expected Frequency:** $E_{ij} = \frac{(Row\ Total) \times (Column\ Total)}{Grand\ Total}$
- **Example Calculation:** Walkthrough an example with a simple 2×2 contingency table.

Conducting the Test

- Step-by-Step Process:
  - Set up the contingency table.
  - Calculate expected frequencies for each cell.
  - Compute Chi-Square statistic using the formula.
  - Compare the calculated Chi-Square to the critical value from Chi-Square distribution table based on degrees of freedom and significance level.

Interpreting Results

- **Critical Values and P-Value:** Use Chi-Square distribution to find P-value or compare the Chi-Square statistic to critical values at common significance levels (e.g., 0.05).
- **Decision Rule:**
  - If χ^2 calculated > χ^2 critical, reject the null hypothesis (significant association).
  - If χ^2 calculated ≤ χ^2 critical, fail to reject the null hypothesis (no significant association).

Example: Product Preference Across Age Groups

- A company wants to determine if there is a significant association between the age group of its customers and their preference for one of two new beverage products. The marketing team has conducted a survey and collected data from different age groups.

Data Collected

- Here's the observed frequency table based on the survey:

	Product A	Product B	Total
Under 30	30	40	70
30 to 50	50	30	80
Over 50	20	30	50
Total	100	100	200

Task

Perform a Chi-Square Test of Independence to determine if there's a significant relationship between age group and product preference.

Steps to Solve in Excel

1. **Prepare the Data in Excel:**
   - Create a new Excel spreadsheet.
   - Enter the data into a worksheet as shown in the table above.
2. **Calculate Expected Frequencies:**
   - Each cell's expected frequency (E) can be calculated using the formula: $E_{ij} = \frac{(Row\ Total) \times (Column\ Total)}{Grand\ Total}$.
   - For instance, the expected frequency for Product A and Under 30 would be $E_{11} = \frac{70 \times 100}{200} = 35$.
3. **Setup Chi-Square Calculation:**
   - Create another table next to the first one to fill in the expected frequencies.
   - Below this table, calculate the Chi-Square statistic using the formula: $\chi^2 = \sum \frac{(O-E)^2}{E}$, where O is the observed frequency and E is the expected frequency.
4. **Use Excel Formula:**
   - For cell A1 (Under 30, Product A): `=((B2-B8)^2)/B8` where B2 is observed and B8 is expected frequency.
   - Drag and fill this formula for all cells.

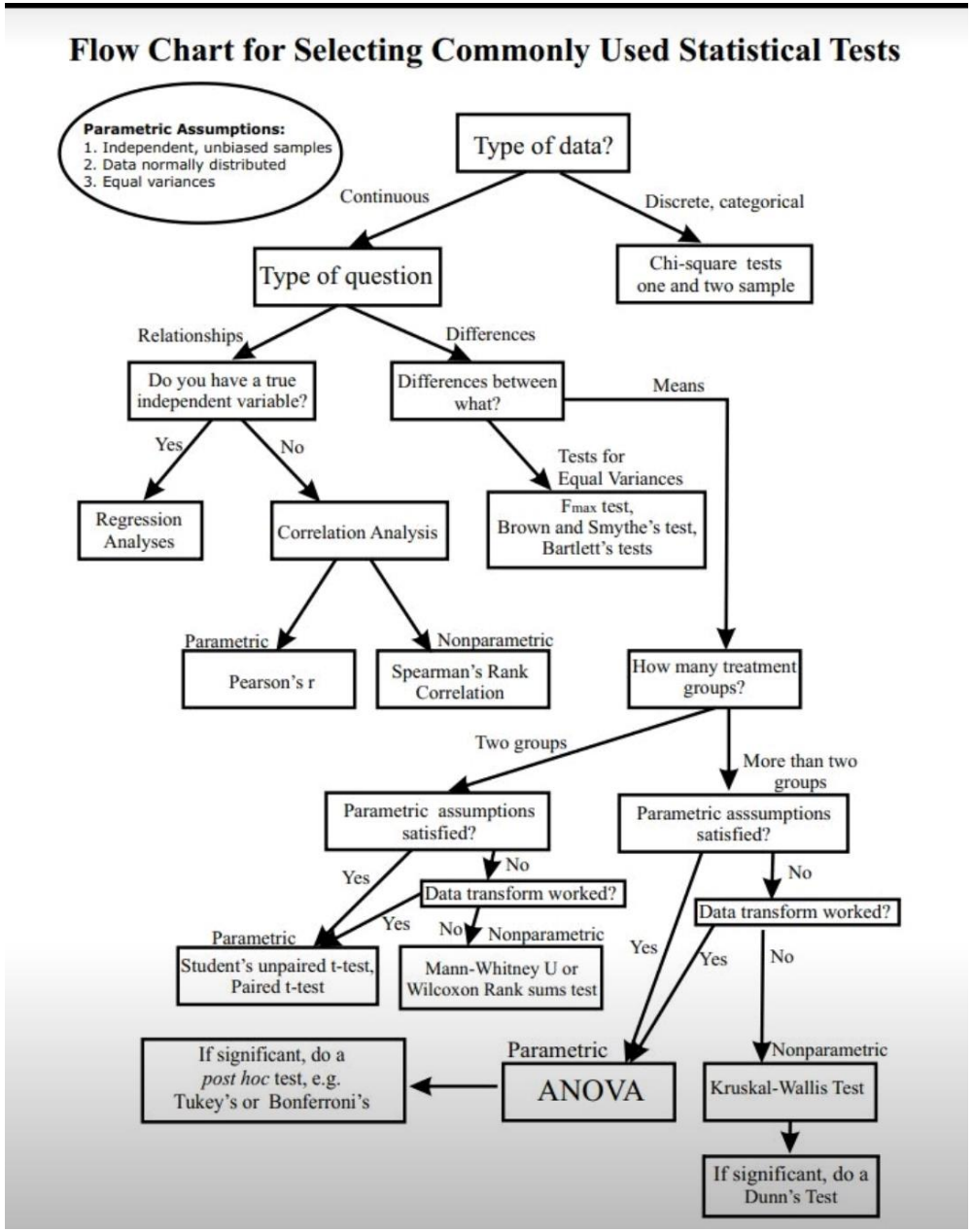
Steps to Solve in Excel

5. **Sum the Results:**
   - Add up all the individual values calculated in the above step to get the total Chi-Square statistic.
6. **Determine the Degrees of Freedom:**
   - Calculate degrees of freedom as $(rows - 1) \times (columns - 1)$. Here, it would be $(3 - 1) \times (2 - 1) = 2$.
7. **Compare to Critical Value:**
   - Use a Chi-Square distribution table or Excel's CHISQ.DIST.RT function to find the critical value or p-value. The formula in Excel would be: `=CHISQ.DIST.RT(chi_square_statistic, degrees_freedom)`
   - Generally, if p-value < 0.05, the result is significant.

Interpretation

- After calculating the Chi-Square statistic and comparing it with the critical value from a Chi-Square distribution table (or using p-value), you can conclude whether there is a statistically significant association between age group and product preference.

How to Select Test Type



Regression Analysis

Introduction to Regression Analysis

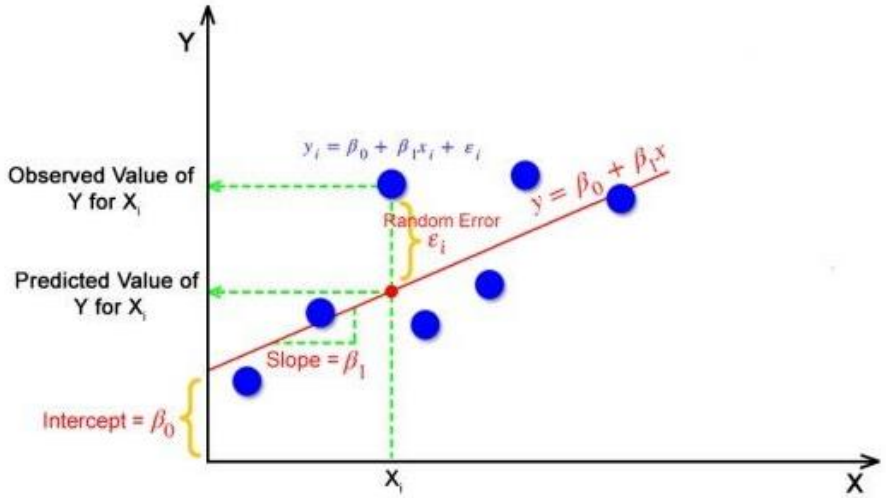
- Definition: Regression analysis is a powerful statistical method used for predicting a dependent variable based on one or more independent variables.
- Purpose: To understand how the typical value of the dependent variable changes when any one of the independent variables is varied, while the other independent variables are held fixed

Types of Regression

- Simple Linear Regression: Involves one independent variable and one dependent variable and the relationship between them is modeled by a linear function.
- Multiple Linear Regression (Focus of this presentation): Involves multiple independent variables influencing a single dependent variable.
- Other Types: Logistic regression, polynomial regression, Ridge Regression, Lasso Regression etc., used for more specific types of data and relationships.

The Regression Equation

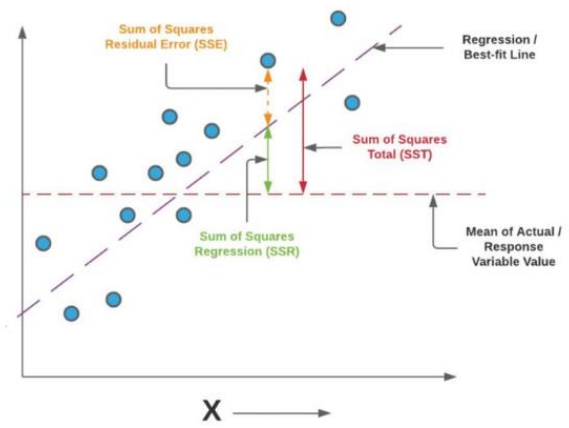
- **Equation:** $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon$
  - Y: Dependent variable
  - β_0: Y-intercept
  - $\beta_1, \beta_2, \ldots, \beta_n$: Coefficients of variables
  - $X_1, X_2, \ldots, X_n$: Independent variables
  - ϵ: Error term



Finding the Best Fit Line

- The goal of the linear regression algorithm is to get the best values for B0 and B1 to find the best fit line.
- The best fit line is a line that has the least error which means the error between predicted values and actual values should be minimum.
- In regression, the difference between the observed value of the dependent variable(Yi) and the predicted value(predicted) is called the residuals.
- *εi = Y Predicted – Yi*
- *Where Y Predicted = B0 + B1 * Xi*
- In simple terms, the best fit line is a line that fits the given scatter plot in the best way. Mathematically, the best fit line is obtained by minimizing the Residual Sum of Squares(RSS).

$$R^2 = \frac{\text{Variance explained by the model}}{\text{Total variance}}$$



Mean Squared Error (MSE)

The MSE cost function is given by:

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2$$

Where:

- $J(\theta)$ is the cost function.
- m is the number of training examples.
- $h_\theta(x^{(i)})$ is the predicted value (hypothesis) for the i-th example.
- $y^{(i)}$ is the actual value for the i-th example.
- $x^{(i)}$ is the feature vector for the i-th example.
- θ represents the parameters (coefficients) of the linear regression model.

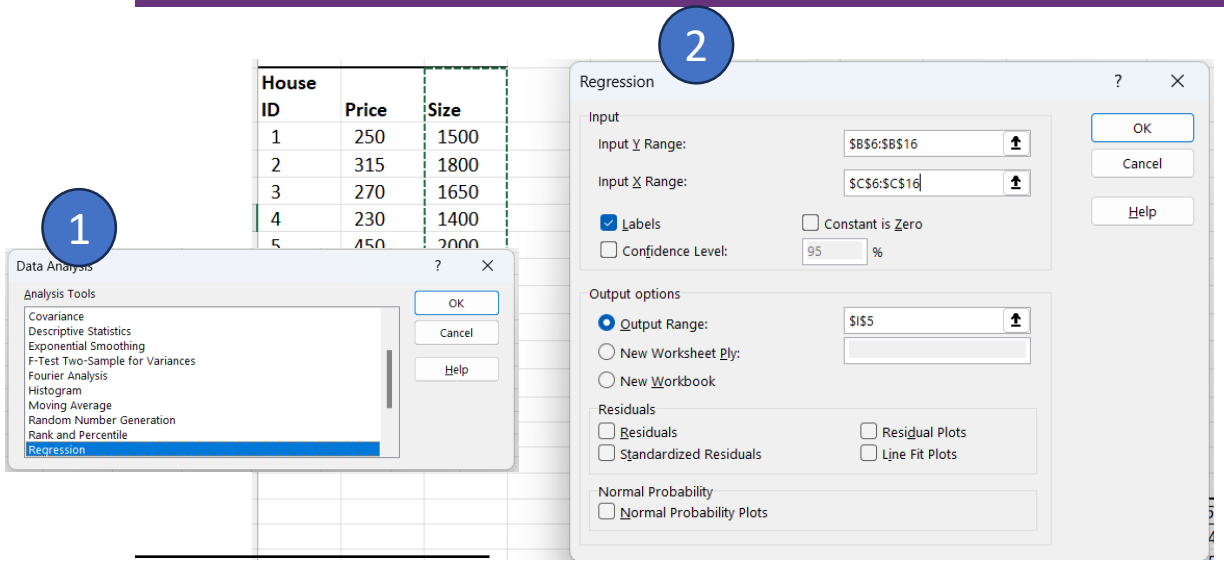
Evaluating the Model

- R-squared (The Goodness of Fit): Proportion of variance in the dependent variable that is predictable from the independent variables.
- Adjusted R-squared: Modified version of R-squared adjusted for the number of predictors.
- F-Statistic: Tests whether at least one predictor variable has a non-zero coefficient.
- t-Tests on Coefficients: Each beta coefficient has an associated t-test to determine if the variable is statistically significant.

Linear Regression Example

- Scenario: A real estate company wants to predict the selling price of houses based on various features.
- Variables:
  - Price: Selling price of the house (in thousands of dollars).
  - Size: Size of the house in square feet.
  - Bedrooms: Number of bedrooms.
  - Age: Age of the house in years.
  - Location: Categorical variable indicating the neighborhood quality (1 = Low, 2 = Medium, 3 = High).

Simple Linear Regression



House ID	Price	Size
1	250	1500
2	315	1800
3	270	1650
4	230	1400
5	450	2000
6	190	1300
7	350	1850
8	300	1600
9	280	1500
10	310	1700

3

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.946288
R Square	**0.895461**
Adjusted R	0.882394
Standard E	24.46997
Observatic	10

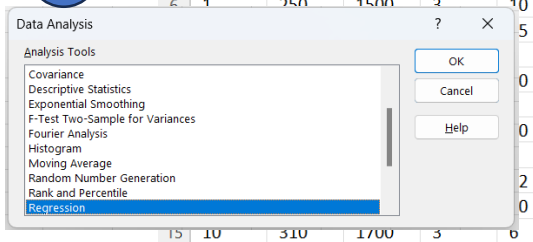
ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regressior	1	41032.27	41032.27	68.52653	3.41185E-05
Residual	8	4790.234	598.7793		
Total	9	45822.5			

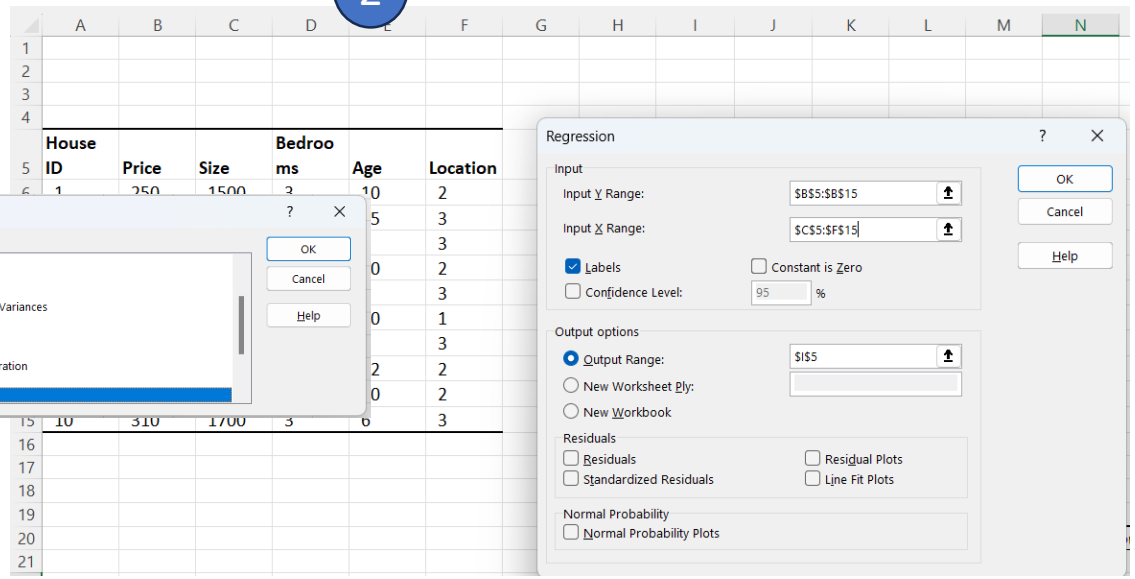
	Coefficients	*andard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	-217.422	62.32294	-3.48863	0.008215	-361.1388328	-73.7049	-361.139	-73.7049
Size	0.314063	0.037939	8.278075	3.41E-05	0.226574839	0.40155	0.226575	0.40155

Multiple Linear Regression

1



2



House ID	Price	Size	Bedrooms	Age	Location
1	250	1500	3	10	2
2	315	1800	4	15	3
3	270	1650	3	5	3
4	230	1400	2	20	2
5	450	2000	4	2	3
6	190	1300	3	30	1
7	350	1850	4	8	3
8	300	1600	3	12	2
9	280	1500	3	10	2
10	310	1700	3	6	3

3

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.990811
R Square	**0.981707**
Adjusted R	0.967073
Standard E	12.94776
Observatic	10

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regressior	4	44984.28	11246.07	67.08281	0.000156
Residual	5	838.2229	167.6446		
Total	9	45822.5			

	Coefficients	*andard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	-322.337	73.17067	-4.40527	0.006987	-510.428	-134.246	-510.428	-134.246
Size	0.56681	0.075552	7.502238	0.000665	0.372597	0.761023	0.372597	0.761023
Bedrooms	-36.5822	14.88472	-2.4577	0.057391	-74.8445	1.680218	-74.8445	1.680218
Age	-1.38428	0.964258	-1.4356	0.210597	-3.86299	1.094419	-3.86299	1.094419
Location	-72.3608	15.29873	-4.72986	0.005197	-111.687	-33.0342	-111.687	-33.0342

Assumptions of Linear Regression

1. Linearity

- The relationship between the independent variables (predictors) and the dependent variable (response) should be linear. This means that the change in the dependent variable is proportional to the change in the independent variable.
- Example: If you are predicting house prices (dependent variable) based on the size of the house (independent variable), a linear relationship implies that each additional square foot of the house size increases the price by a constant amount.

2. Independence

- Observations should be independent of each other. This means that the residuals (errors) should not be correlated across observations. This assumption is important for the validity of standard statistical tests.
- Example: In a study measuring the effect of a new drug on blood pressure, each patient's measurement should be independent. If measurements from the same patient at different times are used, they are not independent.

3. Homoscedasticity

- The variance of the error terms (residuals) should be constant across all levels of the independent variables. If this assumption is violated, it indicates heteroscedasticity, meaning the spread of residuals varies at different levels of the independent variables.
- Example: When plotting residuals against fitted values in a regression analysis of car mileage against engine size, the spread of residuals should be roughly constant. If residuals spread out more as engine size increases, this indicates heteroscedasticity.

Assumptions of Linear Regression

4. Normal Distribution of Errors

- The error terms should be normally distributed. This assumption is especially important for small sample sizes because it affects the validity of confidence intervals and hypothesis tests.
- Example: After fitting a regression model predicting student test scores based on study hours, plotting a histogram or a Q-Q plot of the residuals should show that they follow a normal distribution.

5. No Multicollinearity

- Independent variables should not be too highly correlated with each other. High correlation between predictors (multicollinearity) can inflate the variances of the coefficient estimates and make the model unstable.
- Example: In a regression model predicting salary based on education level and years of experience, if education level and years of experience are highly correlated, it could cause multicollinearity issues. Checking the Variance Inflation Factor (VIF) can help detect this.

Determine Sample Size for Proportion Estimation

Step-by-Step Process:

1. **Determine the Confidence Level:**
   - Common confidence levels are 90%, 95%, and 99%. For most academic and business purposes, a 95% confidence level is typical.
2. **Specify the Margin of Error (ME):**
   - The margin of error is the range within which you expect the true population parameter to lie. Common margins of error are 5% (0.05), 3% (0.03), and 1% (0.01).
3. **Estimate the Population Proportion (p):**
   - If you have a prior estimate of the population proportion, use it. If not, use 0.5 as a conservative estimate, as it maximizes the required sample size.
4. **Use the Sample Size Formula:**
   - For large populations, the sample size (n) can be approximated using the formula:
     $n = \frac{Z^2 \times p \times (1-p)}{ME^2}$
     Where Z is the z-score corresponding to your confidence level.
   - For a 95% confidence level, $Z \approx 1.96$.
   - For a 99% confidence level, $Z \approx 2.576$.

Determine Sample Size for Proportion Estimation

Example Calculation:

Let's calculate the sample size needed for a 95% confidence level, a margin of error of 5%, and an estimated population proportion of 0.5.

1. **Set the Parameters:**

   - Confidence level: 95% (Z = 1.96)
   - Margin of error: 5% (ME = 0.05)
   - Estimated population proportion: $p = 0.5$

2. **Plug the Values into the Formula:**

$$n = \frac{(1.96)^2 \times 0.5 \times (1-0.5)}{(0.05)^2}$$
$$n = \frac{3.8416 \times 0.25}{0.0025}$$
$$n = \frac{0.9604}{0.0025}$$
$$n = 384.16$$

So, you would need a sample size of approximately 385 students.

Determine Sample Size for Proportion Estimation

Adjusting for a Finite Population:

Since your population size (N) is relatively small (600 students), you can adjust the sample size using the finite population correction formula:

$$n_{adj} = \frac{n}{1+\frac{n-1}{N}}$$

Where:

- n is the initial sample size calculated (385).
- N is the population size (600).

$$n_{adj} = \frac{385}{1+\frac{385-1}{600}}$$
$$n_{adj} = \frac{385}{1+\frac{384}{600}}$$
$$n_{adj} = \frac{385}{1+0.64}$$
$$n_{adj} = \frac{385}{1.64}$$
$$n_{adj} \approx 235$$

Final Sample Size:

For a population of 600 students, you would need a sample size of approximately 235 students to achieve a 95% confidence level with a 5% margin of error.